POSITIVITY OF SINGLE-CENTRED BLACK HOLE INDICES IN THE $\mathbb{Z}_2$ CHL MODEL

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Abstract. We prove Sen's positivity conjecture for single-centred quarter-BPS black holes of the $\mathbb{Z}_2$ CHL model for every charge with $Q^2 \leq 2$ and for every charge of discriminant at most 800. For all charges we reduce the conjecture to an explicit level-two Rademacher expansion of the dyon partition function and prove it under that hypothesis.

Keywords: CHL model; single-centred black hole; dyon index; mock Jacobi form; Rademacher expansion.

MSC 2020: 11F50 (primary); 11F37, 11F30, 81T30, 83E30.

PACS: 11.25.-w; 04.70.Dy; 11.25.Mj; 02.10.De.

1. Introduction

A quarter-BPS dyon of $\mathcal{N} = 4$ string theory in four dimensions carries an electric charge Q and a magnetic charge P, and the protected index that counts such states is the helicity trace B_6. When the charges admit a single-centred black hole whose only hair are the zero modes of the broken supersymmetries, the twelve fermion zero modes fix the sign of the trace, and the horizon degrees of freedom contribute $-B_6 > 0$. Sen turned this observation into a conjecture about Fourier coefficients of Siegel modular forms [29]: the index $d^*(Q, P)$ of the single-centred black hole, extracted from the inverse of the dyon partition function with the contour of the attractor region, is positive.[1]

For type II string theory on $K3 \times T^2$, where the partition function is the Igusa cusp form Φ_{10} [16, 31, 19], the conjecture has been settled in two steps. Bringmann and Murthy proved it for the two lowest magnetic charges $P^2 = 2, 4$ by writing the finite parts of the first two Fourier–Jacobi coefficients of $1/\Phi_{10}$ through class numbers [4], using the decomposition into polar and finite parts of Dabholkar, Murthy and Zagier [13]. Rosselló then proved positivity for all charges [24]: he wrote the single-centred index as a mixed Rademacher series, whose existence for $SL_2(\mathbb{Z})$ goes back to Bringmann and Manschot [5] and was used for these black holes by Ferrari and Reys [18], and showed that one term of the series dominates all the others. The CHL models [8] are the $\mathbb{Z}_N$ orbifolds of type IIB on $K3 \times T^2$ in which the $\mathbb{Z}_N$ acts on $K3$ by a symplectic automorphism and on T^2 by a shift; their partition functions $\widetilde{\Phi}_k$ are Siegel modular forms of weight k for subgroups of $Sp(2, \mathbb{Z})$ [22, 12, 15, 14].[2] Bhand, Sen and Singh showed that the Fourier–Jacobi coefficients of $1/\widetilde{\Phi}_k$ are meromorphic Jacobi forms, split them into an Appell–Lerch polar part and a mock Jacobi finite part, and identified the single-centred index with the Fourier coefficients of the finite part on an explicit set of charges

[1] Sen checked the sign on the charges of his Tables 1 to 5, for the heterotic string on T^6 and for the CHL models of orders $2, 3, 5, 7$, and stated the conjecture for all charges in the cones (3.7)–(3.12) of [29]. The entropy function and attractor background behind the statement are reviewed in [28].

[2] For $N = 2$ the model is the heterotic string on T^6 obtained from the $E_8 \times E_8$ string by exchanging the two E_8 factors together with a half shift on a circle [8]; for the ten-dimensional $E_8 \times E_8$ theory and its relation to M-theory see [26]. Background on the involutions of $K3$ surfaces and on Calabi–Yau threefolds fibred by $K3$ surfaces, which underlie the dual type II description, is in [6, 2].

[7, Theorem 5.4]. Cardoso, Nampuri and Rosselló computed the polar coefficients of these finite parts through continued fractions and wrote the immortal index as a Rademacher series, assuming that such an expansion holds at level N [9]; Singh constructed a generating function for the single-centred index [30]. Neither result decides the sign.

This paper treats the $\mathbb{Z}_2$ CHL model. It has $k = 6$, duality group $\Gamma_1(2)$, and charges normalised so that $m = Q^2$, $n = P^2/2$ and $\ell = Q \cdot P$ are integers; the discriminant is $D = Q^2P^2 - (Q \cdot P)^2 = 2mn - \ell^2$. Our results are three.

Theorem A (first two rows)**.** *The single-centred index $d^*(Q, P)$ is positive for every charge with $Q^2 \in \{1, 2\}$, $P^2 > 0$ and $D > 0$, and for every charge in the $\Gamma_1(2)$ orbit of such a charge.*

Theorem B (bounded discriminant)**.** *The single-centred index $d^*(Q, P)$ is positive for every charge with $0 < D \leq 800$.*

Theorem C (all charges)**.** *Assume Hypothesis* R*. Then $d^*(Q, P) > 0$ for every charge with $D > 0$.*

Theorem A has a proof in the style of [4]: a product formula identity puts the Fourier–Jacobi coefficient φ_m of $1/\widetilde{\Phi}_6$ into closed form for $m = 1, 2$, the theorem of Dabholkar, Murthy and Zagier expresses the mock part through Hurwitz class numbers, and an analysis of theta components gives the sign of every coefficient. The new ingredient at $Q^2 = 2$ is the factorisation

$$J_2 = \frac{16\,\theta_4(\sigma, v)^2}{\theta_4(\sigma)^6}\Big[\theta_2(\sigma)^2\big(\theta_3(\sigma)^4 + 3\theta_2(\sigma)^4\big)\theta_2(\sigma, v)^2 - \theta_3(\sigma)^2\big(3\theta_3(\sigma)^4 + \theta_2(\sigma)^4\big)\theta_3(\sigma, v)^2\Big]$$

of the holomorphic part of the finite part (Proposition 4.5), from which the signs of its theta components are manifest.

Theorem B rests on a reduction proved in Section 5. Every charge has, in its duality orbit or after the Fricke exchange of Q^2 and $P^2/2$, a representative with $m \leq n$ and $0 \leq \ell \leq m$, which lies in the set of Bhand, Sen and Singh where d^* is a Fourier coefficient of $1/\widetilde{\Phi}_6$ in a fixed chamber (Proposition 5.6). The Fricke exchange is an exact symmetry of the product formula (Lemma 5.1), a consequence of the identity $\mathfrak{E}(\frac{\tau}{2}) + \mathfrak{E}(\frac{\tau+1}{2}) = 2\mathfrak{E}(\tau)$ between Eisenstein series of level two. For $D \leq 800$ there are finitely many such representatives, and we compute every one of them exactly.

Theorem C follows the strategy of [24], transported to level two. Hypothesis R (Section 7) is an explicit formula for the finite-part coefficients: a vector-valued Rademacher series over the three cosets of $\Gamma^0(2)$ in $SL_2(\mathbb{Z})$, with the principal parts at the two cusps given by sums over Farey walls (the parts (P0) and (P∞)) and with the correction terms coming from the shadow of the mock Jacobi form. For $SL_2(\mathbb{Z})$ such an expansion is a theorem [5]; for the weakly holomorphic part at level two it is Zuckerman's classical formula [36]; for the mixed mock part at level two it has been stated [9] but, to our knowledge, not proved. We have checked it against the exact coefficients to relative precision 10^{-21} at $Q^2 = 1, 2$ and 10^{-13} at $Q^2 = 3, 4$, and we have checked its polar input on every polar coefficient with $Q^2 \leq 10$ at one cusp and $Q^2 \leq 14$ at the other (Section 7.4).

The mechanism behind Theorem C is the following. Write $x = \sqrt{D}/m$ and $\theta = \pi\ell/m$, and let $\mathcal{E}(D, m)$ be the contribution of the most polar term at the cusp 0, an explicit multiple of a Bessel function $I_{15/2}$. After the reduction one has $x \geq 1$. The terms of the Rademacher series attached to the leading family of Farey walls, those with the matrix $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, add up, after division by $\mathcal{E}$, to an approximation of

$$|P(z)|^2, \qquad P(z) = \prod_{n \geq 1}(1 - z^n)^{-8}(1 - z^{2n})^{-8}, \qquad z = e^{-\pi x + i\theta},$$

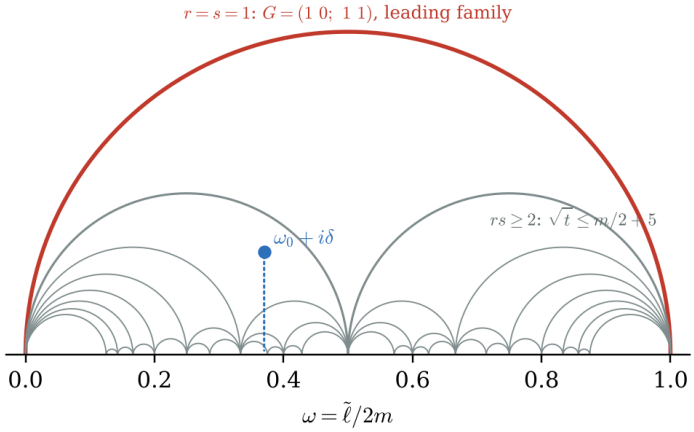


FIGURE 1. The Farey walls of a polar term at the cusp 0, drawn as geodesics from q/s to p/r in the upper half-plane of $\omega = \tilde{\ell}/2m$. A matrix $G = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$ contributes to the polar coefficient at $(\tilde{\ell}, \nu)$ when its geodesic passes over the point $\omega_0 + i\delta$ with $\omega_0 = \tilde{\ell}/2m$ and $\delta = \sqrt{-\tilde{D}}/2m$ (Lemma 6.1). The outer red geodesic is the leading family $r = s = 1$; every other wall lies below height $\frac{1}{4} + \frac{5}{2m}$.

the squared modulus of the generating function of the degeneracies of the electric half-BPS states of the model (Theorem 8.1). The function $|P|^2$ stays above 0.538 on the disc $|z| \leq e^{-\pi}$, its minimum being attained at $z = -e^{-\pi}$ (Proposition 8.3, Figure 3). Everything else, namely the error of the approximation, the remaining walls, the other two cusps of the coset sum, the terms with $\gamma \geq 2$ and the mock corrections, is bounded by 0.347 for $3 \leq Q^2 \leq 100$ by a certificate in interval arithmetic and by 0.3382 for $Q^2 \geq 101$ by explicit inequalities (Theorems 11.1 and 11.2); the margin is plotted in Figure 5.

The heterotic analogue of this picture is the dominance argument of [24], whose model term is a single exponential; the point at level two is that the dominant contribution is not one term but a whole family, and the family resums to an eta quotient. What prevents the argument from being unconditional is only Hypothesis R: every estimate after it is proved.

Section 2 recalls the partition function, the decomposition of Bhand, Sen and Singh and the auxiliary functions. Sections 3 and 4 prove Theorem A. Section 5 proves the reduction and Theorem B. Section 6 studies the Farey walls, Section 7 states Hypothesis R and its evidence, Section 8 proves the eta-quotient picture, Section 9 collects the needed facts on Bessel functions, Section 10 bounds every remaining term, and Section 11 proves Theorem C. Section 12 compares the values with the product formula and with Sen's table. The appendices describe the exact computation and the interval certificates and list the margins.

2. The partition function and the single-centred index

2.1. **Notation.** Write $\mathbf{e}(t) = e^{2\pi i t}$. The variables of the Siegel upper half-space are (ρ, σ, v); put $x = \mathbf{e}(\sigma/2)$, $p = x^2 = \mathbf{e}(\sigma)$ and $y = \mathbf{e}(v)$. For a function of σ alone we use the expansion

in x. The Jacobi theta functions are

$$\theta_1(\tau,z) = -i\sum_{r\in\mathbb{Z}+\frac12}(-1)^{r-\frac12}q^{r^2/2}\zeta^r, \qquad \theta_2(\tau,z) = \sum_{r\in\mathbb{Z}+\frac12} q^{r^2/2}\zeta^r,$$
$$\theta_3(\tau,z) = \sum_{n\in\mathbb{Z}} q^{n^2/2}\zeta^n, \qquad \theta_4(\tau,z) = \sum_{n\in\mathbb{Z}}(-1)^n q^{n^2/2}\zeta^n,$$

with $q=\mathbf{e}(\tau)$, $\zeta=\mathbf{e}(z)$, and $\theta_k(\tau)=\theta_k(\tau,0)$. Let $E(\tau)=1-24\sum\sigma_1(n)q^n$, $E_4=1+240\sum\sigma_3(n)q^n$, $E_6=1-504\sum\sigma_5(n)q^n$, and let

$$\mathfrak{E}(\tau) = 2E(2\tau)-E(\tau) = 1+24\sum_{n\geq1}\sigma_1^{\mathrm{odd}}(n)q^n, \qquad \sigma_1^{\mathrm{odd}}(n)=\sum_{d|n,\ d\ \mathrm{odd}} d,$$

be the Eisenstein series of weight two for $\Gamma_0(2)$. The weak Jacobi forms of weight -2 and 0 and index one are

$$\phi_{-2,1}(\tau,z) = -\frac{\theta_1(\tau,z)^2}{\eta(\tau)^6} = \zeta{-}2{+}\zeta^{-1}{+}O(q), \qquad \phi_{0,1}(\tau,z) = 4\sum_{k=2}^{4}\frac{\theta_k(\tau,z)^2}{\theta_k(\tau)^2} = \zeta{+}10{+}\zeta^{-1}{+}O(q)$$

[17, §9]; we write $A=\phi_{-2,1}(\sigma,v)$ and $B=\phi_{0,1}(\sigma,v)$. The Hurwitz class number $H(D)$ is extended by $H(0)=-\frac{1}{12}$ and $H(D)=0$ unless D is a nonnegative integer with $D\equiv 0,3 \pmod 4$. Put $g(\tau)=\eta(\tau)^8\eta(2\tau)^8$ and

$$\frac{1}{g(\sigma/2)} = \sum_{j\geq0}\beta(j)\,x^{j-1}, \qquad \sum_{j\geq0}\beta(j)x^j = \prod_{r\geq1}(1-x^r)^{-e_r}, \quad e_r=\begin{cases}8, & r \text{ odd},\\ 16, & r \text{ even},\end{cases} \tag{1}$$

so that every $\beta(j)$ is a positive integer.

2.2. **The product formula.** The twisted elliptic genera of $K3$ for the $\mathbb{Z}_2$ symplectic involution are [12]

$$F^{(0,0)}=\phi_{0,1}, \qquad F^{(0,1)}=\tfrac13\phi_{0,1}+\tfrac23\phi_{-2,1}\,\mathfrak{E}(\tau),$$
$$F^{(1,0)}=\tfrac13\phi_{0,1}-\tfrac13\phi_{-2,1}\,\mathfrak{E}(\tfrac{\tau}{2}), \qquad F^{(1,1)}=\tfrac13\phi_{0,1}-\tfrac13\phi_{-2,1}\,\mathfrak{E}(\tfrac{\tau+1}{2}), \tag{2}$$

each a weak Jacobi form of weight 0 and index one for $\Gamma(2)$, and $SL_2(\mathbb{Z})$ permutes them.[3] Write $F^{(r,s)}(\tau,z)=\sum_{b\in\{0,1\}}\sum_{j\in2\mathbb{Z}+b}\sum_{n\in\mathbb{Z}/2}c_b^{(r,s)}(4n-j^2)q^n\zeta^j$. The dyon partition function of the model is [22, 12, 14]

$$\widetilde{\Phi}_6(\rho,\sigma,v) = \mathbf{e}\big(\rho+\tfrac{\sigma}{2}+v\big)\prod_{b=0}^{1}\prod_{r=0}^{1}\prod_{\substack{k'\in\mathbb{Z}+r/2,\ l\in\mathbb{Z},\ j\in2\mathbb{Z}+b\\ k',l\geq0;\ j<0 \text{ if } k'=l=0}}\big(1-\mathbf{e}(k'\sigma+l\rho+jv)\big)^{\sum_{s=0}^{1}(-1)^{sl}c_b^{(r,s)}(4k'l-j^2)}, \tag{3}$$

and the index of the charges (Q,P) is $(-1)^{Q\cdot P+1}$ times the coefficient of $\mathbf{e}(m\rho)x^ny^\ell$ with

$$m=Q^2, \qquad n=P^2/2, \qquad \ell=Q\cdot P, \tag{4}$$

in the expansion of $1/\widetilde{\Phi}_6$ on the appropriate contour [16, 22, 27].[4]

[3] In the normalisation of [12], $F^{(r,s)}=\frac{1}{N}\operatorname{Tr}_{RR,g^r}\big(g^s(-1)^{F_L+F_R}q^{L_0}\bar q^{\bar L_0}y^J\big)$, so that $F^{(0,0)}$ is half the elliptic genus $2\phi_{0,1}$ of $K3$; the constant terms then give the exponents $12,2,8,0$ used below.

[4] The exponent of $\mathbf{e}(\rho)$ is $NQ^2/2=Q^2$, and the power of $x=\mathbf{e}(\sigma/2)$ is $P^2/2$, as in [7, (5.15)].

Expand $1/\widetilde{\Phi}_6 = \sum_{m\geq -1} \varphi_m(\sigma,v)\,\mathbf{e}(m\rho)$. The factors with $l=0$ in (3) have the exponents $\sum_s c_b^{(r,s)}(-j^2)$, equal to $12,2,8,0$ for $(r,|j|)=(0,0),(0,1),(1,0),(1,1)$ and to 0 for $|j|\geq 2$; together with the prefactor they give

$$\varphi_{-1} = \frac{1}{g(\sigma/2)\,A}, \tag{5}$$

since $\phi_{-2,1} = -\theta_1^2/\eta^6$. Taking the logarithm of the factors with $l\geq 1$ and collecting the powers of $\mathbf{e}(\rho)$,

$$\begin{aligned} \frac{1}{\widetilde{\Phi}_6} &= \varphi_{-1}\,\mathbf{e}(-\rho)\exp\Big(\sum_{l\geq 1} L_l(\sigma,v)\,\mathbf{e}(l\rho)\Big), \\ L_l &= \sum_{td=l}\frac{1}{2dt}\sum_{r,s\in\{0,1\}}(-1)^{sd}\sum_{u=0}^{2d-1}(-1)^{ru}F^{(r,s)}\Big(\frac{t\sigma+u}{d},\,tv\Big). \end{aligned} \tag{6}$$

The sum over u keeps exactly the integral powers of x, so each L_l is a series in x whose coefficients are Laurent polynomials in y; in particular L_l is entire in v. Hence

$$g(\tfrac{\sigma}{2})\,A\,\varphi_1 = L_2 + \tfrac{1}{2}L_1^2, \qquad g(\tfrac{\sigma}{2})\,A\,\varphi_2 = L_3 + L_1L_2 + \tfrac{1}{6}L_1^3. \tag{7}$$

Lemma 2.1. *The functions L_1, L_2, L_3 are invariant under $\Gamma(2)$, $\Gamma(4)$, $\Gamma(6)$ respectively, acting on σ, and L_l has the elliptic property of index l in v. Every polynomial in L_1,L_2,L_3 that is homogeneous of total index I is a Jacobi form of weight 0 and index I for $\Gamma(12)$, even in v and weak at every cusp.*

Proof. Let $\gamma = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right)$ and $M = \left(\begin{smallmatrix} t & u \\ 0 & d \end{smallmatrix}\right)$. Then

$$M\gamma M^{-1} = \begin{pmatrix} a+uc/t & \big(tb+u(e-a)-u^2c/t\big)/d \\ dc/t & e-uc/t \end{pmatrix}.$$

For $(t,d)=(1,1)$ and $\gamma\in\Gamma(2)$, for $(t,d)=(1,2),(2,1)$ and $\gamma\in\Gamma(4)$, and for $(t,d)=(1,3),(3,1)$ and $\gamma\in\Gamma(6)$, this matrix lies in $\Gamma(2)$, under which every $F^{(r,s)}$ is invariant. This gives the invariance, and $\Gamma(12)$ is contained in all three groups. Under $v\mapsto v+\lambda\sigma+\mu$ the variable tv of the term (t,d,u) moves by $\lambda d\,\tau'+(t\mu-\lambda u)$ with $\tau'=(t\sigma+u)/d$, and the elliptic property of index one of $F^{(r,s)}$ produces the factor $\mathbf{e}(-\lambda^2 d\tau' - 2\lambda d\,tv) = \mathbf{e}(-\lambda^2 l\sigma - 2\lambda l v)$, which is the elliptic property of index l. Evenness in v is inherited from the genera. For $\gamma\in SL_2(\mathbb{Z})$ one has $M\gamma = \gamma' M'$ with $\gamma'\in SL_2(\mathbb{Z})$ and M' upper triangular of the same determinant, and $SL_2(\mathbb{Z})$ permutes the $F^{(r,s)}$, none of which contains a negative power of q; so every term is weak at every cusp. □

2.3. **Polar part, finite part and the single-centred index.** By [7, Theorem 5.2], φ_m is a meromorphic Jacobi form of weight -6 and index m for $\Gamma_1(2)$ with double poles at $v\in\mathbb{Z}\sigma+\mathbb{Z}$. Its polar part is [7, (5.45)]

$$\varphi_m^P = \frac{\hat f(m)}{g(\sigma/2)}\,\mathcal{A}_{2,m}, \qquad \mathcal{A}_{2,m}(\sigma,v) = \sum_{s\in\mathbb{Z}}\frac{p^{ms^2+s}y^{2ms+1}}{(1-p^s y)^2}, \tag{8}$$

where $\hat f(m)$ is the coefficient of q^m in $1/\big(\eta(\tau)^8\eta(2\tau)^8\big)$, so that $\hat f(-1),\hat f(0),\hat f(1),\hat f(2),\hat f(3) = 1,8,52,256,1122$. The finite part $\varphi_m^F = \varphi_m - \varphi_m^P = \sum c_m^F(n,\ell)x^n y^\ell$ is a mock Jacobi form.

Let $\mathcal{A}$ be the set of charges of [7, (5.46)–(5.47)]: with $N=2$ and the notation (4), the charges with $m,n>0$ and

$$D := Q^2P^2 - (Q\cdot P)^2 = 2mn - \ell^2 > 0, \tag{9}$$

such that

$$\min_{m_1\in 2\mathbb{Z},\ s\in\mathbb{Z},\ m_1\mid s(s+1)} \Big[\big(m_1\ell+2ms+m\big)^2+m_1^2D\Big] \ \geq\ m^2. \tag{10}$$

By [7, Theorem 5.4(a)], for every charge in $\mathcal{A}$,

$$d^*(Q,P)=(-1)^{\ell+1}c_m^F(n,\ell). \tag{11}$$

The single-centred index is invariant under the action [7, (5.17)–(5.19)] of $\Gamma_1(2)$ on the charges.

Lemma 2.2. *Every charge with $Q^2\in\{1,2\}$, $P^2>0$ and $D>0$ lies in $\mathcal{A}$.*

Proof. If $m_1=0$, the condition $m_1\mid s(s+1)$ reads $s(s+1)=0$, and the bracket is $m^2(2s+1)^2=m^2$. If $m_1\neq 0$, then $m_1^2\geq 4\geq m^2$ and $D\geq 1$, so the bracket is at least m^2. □

So for $Q^2\in\{1,2\}$ the theorem is the statement that $(-1)^{\ell+1}c_m^F(n,\ell)>0$ whenever $D>0$.

2.4. **A positivity lemma.** The last step in both cases divides a series of known sign pattern by g.

Lemma 2.3. *Let $f=-1+\sum_{n\geq 1}a_nx^n$ with $a_n>K$ for every n, and let $0<e_r\leq K$ for every $r\geq 1$. Then $f\prod_{r\geq 1}(1-x^r)^{-e_r}$ has positive coefficients at every x^n with $n\geq 1$.*

Proof. Let $f_0=f$ and $f_s=f_{s-1}(1-x^s)^{-e_s}=-1+\sum_n b_n^{(s)}x^n$. We show by induction on s that $b_n^{(s)}>0$ for $n\geq 1$ and $b_n^{(s)}>K$ for $n>s$; since f_s and the full product agree through x^s, this proves the lemma. The case $s=0$ is the hypothesis. Write $e=e_s$ and $b=b^{(s-1)}$. If $s\nmid n$, then $b_n^{(s)}=\sum_k\binom{e+k-1}{k}b_{n-sk}\geq b_n$, all terms being positive. If $n=sm$, then

$$b_{sm}^{(s)}=\sum_{k=0}^{m-1}\binom{e+k-1}{k}b_{s(m-k)}-\binom{e+m-1}{m}>\binom{e+m-1}{m-1}\Big(K-\frac{e}{m}\Big),$$

since every $b_{s(m-k)}$ exceeds K and $\sum_{k=0}^{m-1}\binom{e+k-1}{k}=\binom{e+m-1}{m-1}$. The right side is $K-e\geq 0$ for $m=1$, and at least $(e+1)(K-\frac{e}{2})\geq(e+1)\frac{K}{2}\geq K$ for $m\geq 2$. □

We also use the elementary bound $H(D)\leq D$ for $D\geq 3$: a reduced form (a,b,c) of discriminant $-D$ has $|b|\leq a\leq\sqrt{D/3}$ and is determined by (a,b), so $H(D)\leq\sqrt{D/3}\,(2\sqrt{D/3}+1)\leq D$.

3. The first row: $Q^2=1$

Let $\mathcal{H}(\sigma,v)=\sum_{n,\ell}H(4n-\ell^2)p^ny^\ell$ be the generating function of the Hurwitz class numbers, $\mathcal{B}=\mathcal{A}_{2,1}$, $r_3(D)$ the number of representations of D as a sum of three squares,

$$M(\tau)=\frac{E_4(\tau)-E_4(2\tau)}{240}=\sum_{n\geq 1}\sigma_3^{\mathrm{odd}}(n)q^n,\qquad \sigma_3^{\mathrm{odd}}(n)=\sum_{d\mid n,\ n/d\ \mathrm{odd}}d^3,$$

$$\Theta(\sigma,v)=\sum_{e,\ell}r_3(2e-\ell^2)\,x^ey^\ell.$$

Then $\Theta=\frac{1}{2}\big[\theta_3(\frac{\sigma}{2},v)\theta_3(\frac{\sigma}{2})^3+\theta_4(\frac{\sigma}{2},v)\theta_4(\frac{\sigma}{2})^3\big]$ is the theta series of the lattice D_4, with the Jacobi variable on one coordinate vector, at $\tau=\sigma/2$.[5] Write $\phi_{-2,1}(\tau,z)=\sum\alpha(4a-r^2)q^a\zeta^r$, so that $\alpha(-1)=1$ and $\alpha(0)=-2$.

[5] A vector of D_4 is an integral vector of even coordinate sum; the term $\frac{1}{2}[\cdots]$ selects these, and its norm is $2e$ when the first coordinate is ℓ and the others contribute $2e-\ell^2$.

Proposition 3.1. *In the $\mathbb{Z}_2$ CHL model*

$$g(\tfrac{\sigma}{2})\,\varphi_1 = 52\,\mathcal{B} + 48\,M(\tfrac{\sigma}{2})\,A - \tfrac{20}{3}\,\Theta - 104\,\mathcal{H}. \tag{12}$$

Consequently $c_1^F(n,\ell)$ depends only on $D = 2n-\ell^2$, and

$$\begin{aligned} c_1^F(n,\ell) &= \mathcal{C}(D) := \sum_{j\ge 0}\beta(j)\,\mathcal{N}(D+2-2j),\\ \mathcal{N}(D) &= 48\sum_{t\ge 1}\sigma_3^{\mathrm{odd}}(t)\,\alpha(D-2t) - \tfrac{20}{3}\,r_3(D) - 104\,H(D). \end{aligned} \tag{13}$$

Proof. Step 1. We claim that

$$L_2 + \tfrac{1}{2}L_1^2 = \tfrac{13}{36}\,B^2 - \tfrac{5}{9}\,\mathfrak{E}(\tfrac{\sigma}{2})\,AB + \Big(\tfrac{4}{45}E_4(\tfrac{\sigma}{2}) + \tfrac{19}{180}E_4(\sigma)\Big)A^2. \tag{14}$$

Every term is a Jacobi form of weight 0 and index two for $\Gamma(4)$, even and holomorphic in v and weak at every cusp (Lemma 2.1; $\mathfrak{E}(\sigma/2)$ and $E_4(\sigma/2)$ are modular for $\Gamma^0(2)\supset\Gamma(4)$). Let G be the difference of the two sides and $G=\chi_0(\sigma)+\chi_2(\sigma)v^2+\chi_4(\sigma)v^4+\cdots$. If $\chi_0=\cdots=\chi_{\nu-2}=0$, the lowest order in v of the transformation law shows that χ_ν is a modular form of weight ν for $\Gamma(4)$ [17, §3], holomorphic at the cusps because G is weak. Since $[PSL_2(\mathbb{Z}):\bar\Gamma(4)]=24$ and the width of the cusp ∞ is four, Sturm's theorem [32] shows that such a form vanishes once its coefficients of $x^0,\dots,x^\nu$ do. The coefficients of G, computed from (6) and (2), vanish through x^{48}, so $\chi_0=\chi_2=\chi_4=0$. For each σ the entire function $G(\sigma,\cdot)$ then vanishes to order at least six at $v=0$, while a nonzero function with the elliptic property of index two has exactly four zeros in a period parallelogram [17, Theorem 1.2]. Hence $G=0$.

Step 2. Bringmann and Murthy [4, Section 3] proved

$$\frac{B^2}{A} = 144\,\mathcal{B} + E_4(\sigma)\,A - 288\,\mathcal{H}, \tag{15}$$

and the identity

$$\mathfrak{E}(\tfrac{\sigma}{2})\,B = \tfrac{1}{5}\big(6E_4(\sigma) - E_4(\tfrac{\sigma}{2})\big)A + 12\,\Theta \tag{16}$$

holds because both sides are holomorphic Jacobi forms of weight two and index one for $\Gamma^0(2)$ [17, §7]: the argument of Step 1, with $[PSL_2(\mathbb{Z}):\bar\Gamma^0(2)]=3$, width two, Sturm bound $x^{\lfloor\nu/4\rfloor}$ and two zeros in a period parallelogram, reduces it to the coefficients through x^{48}, which agree.

Step 3. Divide (14) by A and insert (15) and (16). The coefficient of $\mathcal{B}$ is $\frac{13}{36}\cdot 144 = 52$, that of $\mathcal{H}$ is $-\frac{13}{36}\cdot 288 = -104$, that of Θ is $-\frac{5}{9}\cdot 12 = -\frac{20}{3}$, and that of A is

$$\Big(\tfrac{13}{36} - \tfrac{2}{3} + \tfrac{19}{180}\Big)E_4(\sigma) + \Big(\tfrac{1}{9} + \tfrac{4}{45}\Big)E_4(\tfrac{\sigma}{2}) = \tfrac{1}{5}\big(E_4(\tfrac{\sigma}{2}) - E_4(\sigma)\big) = 48\,M(\tfrac{\sigma}{2}).$$

With (7) this is (12). The term $52\mathcal{B}/g(\sigma/2)$ is the polar part (8), and the remaining numerator has the coefficient $\mathcal{N}(2e-\ell^2)$ at $x^e y^\ell$; multiplying by (1) gives (13). □

Lemma 3.2. (i) $\alpha(D)>0$ *for* $D\equiv 3 \pmod 4$, $\alpha(D)<0$ *for* $D\equiv 0 \pmod 4$ *with* $D\ge 0$, *and* $\alpha(D)=0$ *otherwise.*

(ii) $\mathcal{N}(D)=0$ *for* $D<0$, $\mathcal{N}(0)=2$, $\mathcal{N}(D)>0$ *for odd* $D\ge 1$, *and* $-\mathcal{N}(2t)\ge 96\,\sigma_3^{\mathrm{odd}}(t)$ *for* $t\ge 1$.

Proof. (i) With $\vartheta_{1,\mu}(\tau,z) = \sum_{r\equiv\mu\,(2)} q^{r^2/4}\zeta^r$ one has $\phi_{-2,1} = \big[\vartheta_{1,0}(\tau,0)\vartheta_{1,1}(\tau,z) - \vartheta_{1,1}(\tau,0)\vartheta_{1,0}(\tau,z)\big]/\eta(\tau)^6$ [4, (4.1)]. So $\sum_{D\equiv 3\,(4)}\alpha(D)q^{(D+1)/4} = \vartheta_{1,0}(\tau,0)\prod(1-q^n)^{-6}$ and $\sum_{D\equiv 0\,(4)}\alpha(D)q^{D/4} = -q^{-1/4}\vartheta_{1,1}(\tau,0)\prod(1-q^n)^{-6}$, and both right sides have positive coefficients at every exponent ≥ 0.

(ii) For $D<0$ every argument $D-2t$ is at most -3, and $r_3(D)=H(D)=0$. At $D=0$, $\mathcal{N}(0)=-\frac{20}{3}+\frac{104}{12}=2$. For $D=2t$ every $\alpha(2t-2t')$ is ≤ 0 by (i), the term $t'=t$ is $\alpha(0)=-2$, and $r_3, H\ge 0$; this gives $-\mathcal{N}(2t)\ge 96\,\sigma_3^{\rm odd}(t)$. For odd D every $\alpha(D-2t)$ is ≥ 0, and the term $t=(D+1)/2$ is $\alpha(-1)\sigma_3^{\rm odd}(t)\ge t^3$, so the sum in $\mathcal{N}(D)$ is at least $6(D+1)^3$. Further $r_3(D)\le(2\sqrt{D}+1)\max_{k\le D} r_2(k)\le(2\sqrt{D}+1)\,8\sqrt{D}\le 24D$, from $r_2(k)\le 4d(k)\le 8\sqrt{k}$, and $H(D)\le D$. Hence $\mathcal{N}(D)\ge 6(D+1)^3-264D>0$ for odd $D\ge 7$, while $\mathcal{N}(1),\mathcal{N}(3),\mathcal{N}(5)=8,\,296,\,1568$. □

Theorem 3.3. *In the $\mathbb{Z}_2$ CHL model, $d^*(Q,P)>0$ for every charge with $Q^2=1$, $P^2>0$ and $P^2>(Q\cdot P)^2$, and for every charge in the $\Gamma_1(2)$ orbit of such a charge.*

Proof. By Lemma 2.2, (11) and Proposition 3.1, $d^*=(-1)^{\ell+1}\mathcal{C}(D)$ with $D=2n-\ell^2\equiv\ell \pmod 2$. If D is odd, every argument of $\mathcal{N}$ in (13) is odd, so every term is nonnegative by Lemma 3.2, and the term $j=0$ is $\mathcal{N}(D+2)>0$. If $D=2u$ is even, $-\mathcal{C}(2u)$ is the coefficient of x^{u+1} in $f\prod_r(1-x^r)^{-e_r}$ with $f=\sum_{t\ge 0}\big(-\mathcal{N}(2t)\big)x^t$. By Lemma 3.2, $f/2=-1+\sum_{t\ge1}a_t x^t$ with $a_t\ge 48$, and Lemma 2.3 with $K=16$ gives $-\mathcal{C}(2u)>0$. The orbits follow from the $\Gamma_1(2)$ invariance of d^*. □

4. The second row: $Q^2=2$

4.1. **The closed form.** Let

$$f_6=\tfrac{5}{189}E_6(\tfrac{\sigma}{2})-\tfrac{68}{189}E_6(\sigma),\qquad f_4=\tfrac{4}{45}E_4(\tfrac{\sigma}{2})+\tfrac{26}{45}E_4(\sigma),\qquad f_2=-\tfrac{1}{3}\mathfrak{E}(\tfrac{\sigma}{2}),\tag{17}$$

and $J_2=f_6A^2+f_4AB+f_2B^2$, a weak Jacobi form of weight 2 and index 2 for $\Gamma^0(2)$. Let $\mathcal{H}|V_2$ be the image of $\mathcal{H}$ under the Hecke-type operator V_2 of [13, (4.37)],

$$\mathcal{H}|V_2=\sum_{n\ge0}\sum_{\ell\in\mathbb{Z}}C_2(8n-\ell^2)\,p^n y^\ell,\qquad C_2(\Delta)=H(\Delta)+2H(\Delta/4).\tag{18}$$

Proposition 4.1.

$$L_3+L_1L_2+\tfrac{1}{6}L_1^3=\tfrac{4}{27}\big(B^3-3E_4(\sigma)A^2B+2E_6(\sigma)A^3\big)+A\,J_2.\tag{19}$$

Proof. By Lemma 2.1 both sides are Jacobi forms of weight 0 and index three for $\Gamma(12)$, even and holomorphic in v and weak at every cusp; the functions $E_k(\sigma/2)$ and $\mathfrak{E}(\sigma/2)$ are modular for $\Gamma^0(2)\supset\Gamma(12)$. Let G be the difference and χ_ν its Taylor coefficients in v, as in the proof of Proposition 3.1. Now $[PSL_2(\mathbb{Z}):\bar\Gamma(12)]=576$ and the width of ∞ is 12, so by Sturm's theorem a modular form of weight ν for $\Gamma(12)$ vanishes once its coefficients vanish through $x^{8\nu}$. The coefficients of G vanish through x^{49} (we computed them in exact arithmetic for every monomial $x^e y^\ell$ with $e\le 49$ and $|\ell|\le 18$, which contains every monomial of these orders that a weak form of index three can carry), so $\chi_0=\chi_2=\chi_4=\chi_6=0$. Then $G(\sigma,\cdot)$ vanishes to order at least eight at $v=0$, while a nonzero function with the elliptic property of index three has six zeros in a period parallelogram [17, Theorem 1.2]. Hence $G=0$. □

Proposition 4.2. *In the $\mathbb{Z}_2$ CHL model*

$$g(\tfrac{\sigma}{2})\,\varphi_2=256\,\mathcal{A}_{2,2}+J_2-256\,\mathcal{H}|V_2,\tag{20}$$

so that $g(\frac{\sigma}{2})\varphi_2^F=J_2-256\,\mathcal{H}|V_2$.

Proof. Dabholkar, Murthy and Zagier [13, (9.11)] define

$$\varphi_{2,2}^{\rm opt}=\frac{B^3-3E_4A^2B+2E_6A^3}{1728\,A},$$

a meromorphic Jacobi form of weight 2 and index 2, and prove that its finite part $\varphi_{2,2}^{\mathrm{opt}} - \mathcal{A}_{2,2}$ equals $-\mathcal{H}|V_2$ [13, (9.13) and Theorem 9.1, proved in §10]. Divide (19) by A and use (7): since $\frac{4}{27}\cdot 1728 = 256$,

$$g(\tfrac{\sigma}{2})\,\varphi_2 = 256\,\varphi_{2,2}^{\mathrm{opt}} + J_2 = 256\,\mathcal{A}_{2,2} - 256\,\mathcal{H}|V_2 + J_2.$$

The term $256\mathcal{A}_{2,2}/g(\sigma/2)$ is the polar part (8), as $\hat{f}(2) = 256$. □

Remark 4.3. The index-two weak forms of weight two for $SL_2(\mathbb{Z})$ are spanned by E_6A^2 and E_4AB, whose coefficients at the discriminants $-4, -1$ are $(1, -4)$ and $(1, 8)$; so a weak form of this space with no polar terms vanishes. This is why the optimal choice of [13] is forced at index two, and why the whole of the level-two dependence of φ_2^F sits in the holomorphic form J_2.

4.2. **The factorisation of J_2.** Put $u = \theta_2(\sigma)^4$ and $w = \theta_4(\sigma)^4$, so that $\theta_3(\sigma)^4 = u + w$ by Jacobi's identity. The classical identities

$$\begin{aligned} E_4(\sigma) &= u^2 + uw + w^2, \quad E_4(\tfrac{\sigma}{2}) = 16u^2 + 16uw + w^2, \quad \mathfrak{E}(\tfrac{\sigma}{2}) = 2u + w, \\ E_6(\sigma) &= w^3 + \tfrac{3}{2}uw^2 - \tfrac{3}{2}u^2w - u^3, \quad E_6(\tfrac{\sigma}{2}) = w^3 - 30uw^2 - 96u^2w - 64u^3 \end{aligned} \tag{21}$$

hold between modular forms for $\Gamma(4)$ of weight at most six; by Sturm's theorem each reduces to the first coefficients, which agree [35].[6] From (17),

$$f_2 = -\tfrac{1}{3}(2u + w), \qquad f_4 = 2u^2 + 2uw + \tfrac{2}{3}w^2, \qquad f_6 = -\tfrac{4}{3}u^3 - 2u^2w - \tfrac{4}{3}uw^2 - \tfrac{1}{3}w^3. \tag{22}$$

Lemma 4.4. *For $k = 2, 3, 4$,*

$$Y_k := \frac{\theta_k(\sigma, v)^2}{\theta_k(\sigma)^2} = \frac{B}{12} - e_k A, \qquad e_2 = -\frac{u + 2w}{12}, \quad e_3 = \frac{w - u}{12}, \quad e_4 = \frac{2u + w}{12}.$$

Proof. The quartic relations of Jacobi [34, §21.2],

$$\theta_1(\sigma, v)^2\theta_4^2 = \theta_3(\sigma, v)^2\theta_2^2 - \theta_2(\sigma, v)^2\theta_3^2, \qquad \theta_1(\sigma, v)^2\theta_3^2 = \theta_4(\sigma, v)^2\theta_2^2 - \theta_2(\sigma, v)^2\theta_4^2,$$

with $\theta_k = \theta_k(\sigma)$, and $\eta^3 = \frac{1}{2}\theta_2\theta_3\theta_4$ give $Y_3 - Y_2 = \theta_1(\sigma, v)^2\theta_4^2/(\theta_2^2\theta_3^2) = -\frac{1}{4}\theta_4^4\,A$ and $Y_4 - Y_2 = -\frac{1}{4}\theta_3^4 A$, that is, $Y_3 - Y_2 = (e_2 - e_3)A$ and $Y_4 - Y_2 = (e_2 - e_4)A$. Together with $Y_2 + Y_3 + Y_4 = B/4$ and $e_2 + e_3 + e_4 = 0$ this gives the lemma. □

Proposition 4.5.

$$J_2 = \frac{16\,Y_4}{w}\Big[u(4u + w)\,Y_2 - (u + w)(4u + 3w)\,Y_3\Big]. \tag{23}$$

Equivalently, with $\theta_k = \theta_k(\sigma)$,

$$J_2 = \frac{16\,\theta_4(\sigma, v)^2}{\theta_4^6}\Big[\theta_2^2\big(\theta_3^4 + 3\theta_2^4\big)\theta_2(\sigma, v)^2 - \theta_3^2\big(3\theta_3^4 + \theta_2^4\big)\theta_3(\sigma, v)^2\Big].$$

Proof. By Lemma 4.4, $B = 12(Y_4 + e_4A)$. Substituting into $J_2 = f_6A^2 + f_4AB + f_2B^2$,

$$J_2 = \big(f_6 + 12e_4f_4 + 144e_4^2f_2\big)A^2 + \big(12f_4 + 288e_4f_2\big)AY_4 + 144f_2Y_4^2.$$

By (22) the first bracket vanishes: its coefficients at u^3, u^2w, uw^2, w^3 are $-\frac{4}{3} + 4 - \frac{8}{3}$, $-2 + 6 - 4$, $-\frac{4}{3} + \frac{10}{3} - 2$ and $-\frac{1}{3} + \frac{2}{3} - \frac{1}{3}$. The second bracket is $24u^2 + 24uw + 8w^2 - 8(2u + w)^2 = -8u(u + w)$, and $144f_2 = -48(2u + w)$. So

$$J_2 = -8\,Y_4\big[u(u + w)A + 6(2u + w)Y_4\big].$$

[6] We checked each identity to x^{100} in exact arithmetic, far beyond the Sturm bound x^{12} for weight six on $\Gamma(4)$.

By Lemma 4.4, $A = 4(Y_2 - Y_3)/w$ and $Y_4 = \big((u+w)Y_3 - uY_2\big)/w$. Inserting these, the coefficient of Y_2 in the bracket is $\big(4u(u+w) - 6u(2u+w)\big)/w = -2u(4u+w)/w$ and that of Y_3 is $\big(-4u(u+w) + 6(2u+w)(u+w)\big)/w = 2(u+w)(4u+3w)/w$, which gives (23). The second form follows from $\theta_k^2\theta_k(\sigma,v)^2 = \theta_k^4 Y_k$, $\theta_2^4 = u$, $\theta_3^4 = u+w$ and $\theta_4^4 = w$. □

4.3. **Theta components.** A holomorphic Jacobi form ϕ of index two is determined by its theta components: $\phi = \sum_{\mu \bmod 4} G_\mu(\sigma)\,\vartheta_{2,\mu}(\sigma,v)$ with $\vartheta_{2,\mu}(\sigma,v) = \sum_{\ell\equiv\mu\,(4)} p^{\ell^2/8} y^\ell$ [17, §5]. Since J_2 is even, $G_1 = G_3$. Put

$$a_4 = \theta_3(4\sigma) = \sum_n x^{4n^2}, \qquad b_4 = \theta_2(4\sigma) = \sum_n x^{(2n+1)^2},$$
$$a_2 = \theta_3(2\sigma), \qquad b_2 = \theta_2(2\sigma), \qquad \psi = \sum_{j\in\mathbb{Z}} x^{(4j+1)^2/4}.$$

Then $\theta_3(\sigma) = a_4 + b_4$, $\theta_4(\sigma) = a_4 - b_4$, $a_2^2 = a_4^2 + b_4^2$, $b_2^2 = 2a_4b_4$ and $\theta_2(\sigma)^2 = 2a_2b_2$.

Lemma 4.6. *The theta components of J_2 are*

$$G_0 = -\frac{16\,a_4\big(3a_4^6 + 28a_4^5b_4 + 3a_4^4b_4^2 + 112a_4^3b_4^3 + 25a_4^2b_4^4 + 84a_4b_4^5 + b_4^6\big)}{(a_4-b_4)^4},$$
$$G_2 = -\frac{16\,b_4\big(a_4^6 + 84a_4^5b_4 + 25a_4^4b_4^2 + 112a_4^3b_4^3 + 3a_4^2b_4^4 + 28a_4b_4^5 + 3b_4^6\big)}{(a_4-b_4)^4},$$
$$G_1 = G_3 = \frac{16\,\psi\,\theta_2(\sigma)^2\big(a_4^4 + 28a_4^3b_4 + 6a_4^2b_4^2 + 28a_4b_4^3 + b_4^4\big)}{(a_4-b_4)^4}.$$

In particular $-G_0$, $-G_2$ and G_1 have nonnegative coefficients as series in $x^{1/4}$.

Proof. Expanding the squares, with $\vartheta_{1,\mu} = \vartheta_{1,\mu}(\sigma,v) = \sum_{r\equiv\mu\,(2)} p^{r^2/4} y^r$,

$$\theta_3(\sigma,v)^2 = a_2\vartheta_{1,0} + b_2\vartheta_{1,1}, \qquad \theta_4(\sigma,v)^2 = a_2\vartheta_{1,0} - b_2\vartheta_{1,1}, \qquad \theta_2(\sigma,v)^2 = b_2\vartheta_{1,0} + a_2\vartheta_{1,1}:$$

for the first, write $\frac12(k^2+k'^2) = \frac14(r^2+s^2)$ with $r = k+k'$, $s = k-k' \equiv r \pmod 2$, and sum over s; the others are the same computation with the sign $(-1)^r$ or with $k,k' \in \mathbb{Z}+\frac12$. In the same way

$$\vartheta_{1,0}^2 = a_4\vartheta_{2,0} + b_4\vartheta_{2,2}, \qquad \vartheta_{1,1}^2 = b_4\vartheta_{2,0} + a_4\vartheta_{2,2}, \qquad \vartheta_{1,0}\vartheta_{1,1} = \psi\,(\vartheta_{2,1} + \vartheta_{2,3}).$$

Hence

$$\theta_4(\sigma,v)^2\theta_2(\sigma,v)^2 = a_2b_2(a_4-b_4)(\vartheta_{2,0} - \vartheta_{2,2}) + (a_4-b_4)^2\psi\,(\vartheta_{2,1} + \vartheta_{2,3}),$$
$$\theta_4(\sigma,v)^2\theta_3(\sigma,v)^2 = a_4(a_4-b_4)(a_4+b_4)\,\vartheta_{2,0} - b_4(a_4-b_4)(a_4+b_4)\,\vartheta_{2,2}.$$

Insert these in the second form of Proposition 4.5, with $\theta_2(\sigma)^2 = 2a_2b_2$, $\theta_2^4 = 8a_4b_4(a_4^2+b_4^2)$, $\theta_3^4 = (a_4+b_4)^4$ and $\theta_4^4 = (a_4-b_4)^4$. The coefficient of $\vartheta_{2,1}$ is $16\psi\theta_2(\sigma)^2(\theta_3^4 + 3\theta_2^4)/\theta_4^4$, which is the stated G_1. The coefficients of $\vartheta_{2,0}$ and $\vartheta_{2,2}$ are

$$\frac{16\big[2a_2^2b_2^2(\theta_3^4 + 3\theta_2^4) - a_4(a_4+b_4)^3(3\theta_3^4 + \theta_2^4)\big]}{(a_4-b_4)^5},$$
$$\frac{16\big[b_4(a_4+b_4)^3(3\theta_3^4 + \theta_2^4) - 2a_2^2b_2^2(\theta_3^4 + 3\theta_2^4)\big]}{(a_4-b_4)^5},$$

and expanding the numerators and cancelling one factor $a_4 - b_4$ gives G_0 and G_2. Finally a_4, b_4, a_2, b_2, ψ have nonnegative coefficients, and

$$\frac{1}{\theta_4(\sigma)} = \frac{1}{a_4 - b_4} = \prod_{n\geq 1} \frac{1+x^n}{1-x^n}$$

has positive coefficients, which gives the sign statement. □

Write $G_\mu = \sum_D g_\mu(D)\, x^{D/4}$, so that $J_2 = \sum_{e,\ell} g_{\ell \bmod 4}(4e-\ell^2)\, x^e y^\ell$. In the same way $\mathcal{H}|V_2$ has the coefficient $C_2(4e-\ell^2)$ at $x^e y^\ell$ when e is even and 0 when e is odd. Since $\ell^2 \equiv 0, 4, 1 \pmod 8$ for $\ell \equiv 0, 2, \pm 1 \pmod 4$, the parity of e is determined by $D = 4e - \ell^2$ and $\mu = \ell \bmod 4$. So the numerator $J_2 - 256\,\mathcal{H}|V_2$ of φ_2^F has the coefficient

$$N_\mu(D) = g_\mu(D) - 256\,\epsilon_\mu(D)\, C_2(D), \qquad \epsilon_\mu(D) = \begin{cases} 1, & (D+\mu^2)/4 \text{ even}, \\ 0, & \text{otherwise}, \end{cases} \tag{24}$$

at $x^e y^\ell$, with $D = 4e - \ell^2$ and $\mu \equiv \ell \pmod 4$, $N_3 = N_1$. By Proposition 4.2 and (1),

$$c_2^F(n,\ell) = \sum_{j\geq 0} \beta(j)\, N_{\ell \bmod 4}(D + 4 - 4j), \qquad D = 4n - \ell^2. \tag{25}$$

For $Q^2 = 2$ this D is $Q^2P^2 - (Q\cdot P)^2$, in agreement with (9).

Lemma 4.7. (i) $N_\mu(D) = 0$ *for* $D < 0$.
(ii) $N_1(D) > 0$ *for every* $D \equiv 3 \pmod 4$, $D \geq 3$.
(iii) $N_2(0) = 0$ *and* $N_2(D) < 0$ *for* $D \equiv 0 \pmod 4$, $D \geq 4$.
(iv) $N_0(0) = 16$ *and* $-N_0(4t) \geq 1088$ *for* $t \geq 1$.

Proof. We use $a_4 = 1 + O(x^4)$, $b_4 = 2x + O(x^9)$, $\psi = x^{1/4} + O(x^{9/4})$, $\theta_2(\sigma)^2 = 4x^{1/2} + O(x^{5/2})$, all with nonnegative coefficients, and $1/(a_4 - b_4)^4 \geq 1/(1-x)^4$ coefficientwise. Here $F \geq F'$ means that $F - F'$ has nonnegative coefficients.

(i) By Lemma 4.6, G_0, G_1, G_2 begin with x^0, $x^{3/4}$, x^1, so $g_\mu(D) = 0$ for $D < 0$; and $C_2(D) = 0$ for $D < 0$.

(ii) For $D \equiv 3 \pmod 4$, $D/4$ is not an integer, so $C_2(D) = H(D)$. Keeping the terms a_4^4 and $28a_4^3 b_4$ of the numerator of G_1,

$$G_1 \;\geq\; 64\, x^{3/4}\Big(\frac{1}{(1-x)^4} + \frac{56x}{(1-x)^4}\Big), \qquad g_1(4k+3) \;\geq\; 64\Big[\binom{k+3}{3} + 56\binom{k+2}{3}\Big].$$

At $D = 3$, $(D+1)/4 = 1$ is odd, so $\epsilon_1(3) = 0$ and $N_1(3) = g_1(3) = 64$. For $k \geq 1$, $g_1(4k+3) \geq 3584k$, while $256\, H(4k+3) \leq 256(4k+3) < 3584k$; so $N_1(4k+3) > 0$.

(iii) At $D = 0$, $g_2(0) = 0$ and $\epsilon_2(0) = 0$ because $(0+4)/4 = 1$ is odd. For $D = 4t$, $t \geq 1$, keeping the term a_4^6 of the numerator of G_2 gives $-G_2 \geq 32x/(1-x)^4$, so $-g_2(4t) \geq 32\binom{t+2}{3} > 0$, and $-256\,\epsilon_2 C_2(4t) \leq 0$ because $C_2(D) \geq 0$ for $D > 0$.

(iv) At $D = 0$, $g_0(0) = -48$ and $\epsilon_0(0) = 1$, $C_2(0) = 3H(0) = -\frac{1}{4}$, so $N_0(0) = -48 + 64 = 16$. For $t \geq 1$, keeping $3a_4^6$ and $28a_4^5 b_4$ in the numerator of G_0,

$$-G_0 \;\geq\; \frac{48 + 896x}{(1-x)^4}, \qquad -g_0(4t) \;\geq\; 48\binom{t+3}{3} + 896\binom{t+2}{3} \;\geq\; 1088,$$

and $-N_0(4t) = -g_0(4t) + 256\,\epsilon_0 C_2(4t) \geq -g_0(4t)$. □

The first values are $N_0(0), N_0(4), N_0(8) = 16, -1280, -9536$; $N_2(4), N_2(8) = -160, -5632$; $N_1(3), N_1(7), N_1(11) = 64, 3840, 32960$. Figure 2 shows the pattern of Lemma 4.7.

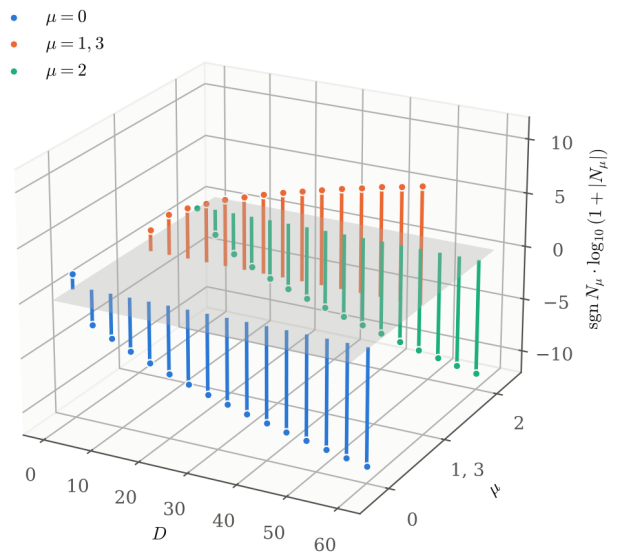


FIGURE 2. The coefficients $N_\mu(D)$ of the numerator of φ_2^F for $0 \le D \le 60$, plotted as $\operatorname{sgn} N_\mu \cdot \log_{10}(1+|N_\mu|)$ over the plane $N = 0$. The component $\mu = 0$ is positive only at $D = 0$, the component $\mu = 2$ is never positive, and the odd components are always positive.

Theorem 4.8. *In the $\mathbb{Z}_2$ CHL model, $d^*(Q,P) > 0$ for every charge with $Q^2 = 2$, $P^2 > 0$ and $2P^2 > (Q \cdot P)^2$, and for every charge in the $\Gamma_1(2)$ orbit of such a charge.*

Proof. By Lemma 2.2 and (11), $d^* = (-1)^{\ell+1} c_2^F(n,\ell)$, and c_2^F is given by (25) with $D = 4n - \ell^2 > 0$; every $\beta(j)$ is positive.

If ℓ is odd, then $D \equiv 3 \pmod 4$, every argument $D + 4 - 4j$ is $\equiv 3 \pmod 4$, every term is nonnegative by Lemma 4.7(i), (ii), and the term $j = 0$ is $N_1(D+4) > 0$. So $d^* = c_2^F > 0$.

If $\ell \equiv 2 \pmod 4$, then $D \equiv 0 \pmod 4$, every term is ≤ 0 by Lemma 4.7(i), (iii), and the term $j = 0$ is $N_2(D+4) < 0$ since $D + 4 \ge 8$. So $d^* = -c_2^F > 0$.

If $\ell \equiv 0 \pmod 4$, write $D = 4u$ with $u \ge 1$. Then $-c_2^F$ is the coefficient of x^{u+1} in $f \prod_r (1-x^r)^{-e_r}$, with $f = \sum_{t \ge 0} \big(- N_0(4t) \big) x^t$. By Lemma 4.7(iv), $f/16 = -1 + \sum_{t \ge 1} a_t x^t$ with $a_t \ge 68 > 16$, and Lemma 2.3 with $K = 16$ gives $-c_2^F > 0$, so $d^* > 0$. The orbits follow from the $\Gamma_1(2)$ invariance of d^*. □

5. Reduction to a fundamental set of charges

From now on m is arbitrary. This section shows that the sign of d^* has to be checked only on charges with $m \le n$ and $0 \le \ell \le m$, and proves Theorem B.

5.1. **The Fricke symmetry of the product formula.** The duality group $\Gamma_1(2)$ acts on the charges by $(Q,P) \mapsto (aQ + bP,\, cQ + dP)$ with $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_1(2)$, that is, c even and a, d odd [7, (5.17)–(5.19)]. It does not exchange Q^2 with $P^2/2$. The product formula has, however, an extra symmetry that does.

Lemma 5.1. *Let $E(A,B,j) = \sum_{s=0}^{1} (-1)^{sB} c_{j \bmod 2}^{(A \bmod 2,\, s)} (2AB - j^2)$ be the exponent of the factor $1 - x^A \mathbf{e}(B\rho) y^j$ in (3), where $x = \mathbf{e}(\sigma/2)$ and $A = 2k'$. Then $E(A,B,j) = E(B,A,j)$ for all*

$A, B \geq 0$ and j. Consequently

$$\widetilde{\Phi}_6(\rho,\sigma,v) = \widetilde{\Phi}_6\big(\tfrac{\sigma}{2},\, 2\rho,\, v\big) \tag{26}$$

as formal products.

Proof. If A and B have the same parity, the two sides are the same expression. Let A be even and B odd; then $2AB - j^2 \equiv -j^2 \pmod 4$ and $AB/2 \in \mathbb{Z}$, and

$$E(A,B,j) = c^{(0,0)}(N) - c^{(0,1)}(N), \qquad E(B,A,j) = c^{(1,0)}(N) + c^{(1,1)}(N), \qquad N = 2AB - j^2,$$

both read off at an integral power of q. By (2),

$$F^{(0,0)} - F^{(0,1)} = \tfrac{2}{3}\phi_{0,1} - \tfrac{2}{3}\phi_{-2,1}\,\mathfrak{E}(\tau), \qquad F^{(1,0)} + F^{(1,1)} = \tfrac{2}{3}\phi_{0,1} - \tfrac{1}{3}\phi_{-2,1}\big(\mathfrak{E}(\tfrac{\tau}{2}) + \mathfrak{E}(\tfrac{\tau+1}{2})\big).$$

Since $\mathfrak{E}(\tau) = 1 + 24\sum \sigma_1^{\mathrm{odd}}(n) q^n$ and $\sigma_1^{\mathrm{odd}}(2k) = \sigma_1^{\mathrm{odd}}(k)$,

$$\mathfrak{E}(\tfrac{\tau}{2}) + \mathfrak{E}(\tfrac{\tau+1}{2}) = 2 + 48\sum_{k\geq 1} \sigma_1^{\mathrm{odd}}(2k) q^k = 2\mathfrak{E}(\tau),$$

so $F^{(0,0)} - F^{(0,1)} = F^{(1,0)} + F^{(1,1)}$ identically, and the exponents agree. The prefactor $\mathbf{e}(\rho + \frac{\sigma}{2} + v)$ and the range $A, B \geq 0$, $j < 0$ if $A = B = 0$, of the product are symmetric under $A \leftrightarrow B$, which gives (26). □

Remark 5.2. The substitution $(\rho,\sigma) \mapsto (\sigma/2, 2\rho)$ is the action on the Siegel upper half-space of the Fricke involution of $\Gamma_0(2)$; Persson and Volpato showed that the CHL models with a geometric symmetry are self-dual under the corresponding Fricke S-duality [23], and (26) is its shadow on the partition function. We use only the elementary statement of Lemma 5.1.

5.2. **A chamber shared by both orientations.** For a charge (m,n,ℓ) and a point (ρ_2,σ_2,v_2) of imaginary parts, let

$$c(m,n,\ell;\rho_2,\sigma_2,v_2) = \int \frac{\mathbf{e}(-m\rho - n\frac{\sigma}{2} - \ell v)}{\widetilde{\Phi}_6(\rho,\sigma,v)}\, d\rho_1\, d\sigma_1\, dv_1,$$

the integral running over a period cell of the real parts. Sen showed [27, §4] that this number is locally constant in the imaginary parts and jumps only across the walls of marginal stability, which for the $\mathbb{Z}_2$ CHL model are the hyperplanes

$$j\,v_2 + n_1\sigma_2 - m_1\rho_2 = 0, \qquad j \text{ odd}, \quad m_1 \in \mathbb{Z}, \quad n_1 \in \tfrac{1}{2}\mathbb{Z}, \quad m_1 n_1 = \tfrac{1-j^2}{4}, \tag{27}$$

coming from the double zeros of $\widetilde{\Phi}_6$ with vanishing coefficient of $\rho\sigma - v^2$.

Lemma 5.3. *Let $\mathcal{R} = \{\rho_2, \sigma_2 > 0,\ 0 < -v_2 < \frac{1}{2}\min(\rho_2, \frac{\sigma_2}{2})\}$.*

(i) *$\mathcal{R}$ meets no wall (27), so $c_{\mathcal{R}}(m,n,\ell) = c(m,n,\ell;\rho_2,\sigma_2,v_2)$ does not depend on the point of $\mathcal{R}$.*
(ii) *$c_{\mathcal{R}}(m,n,\ell) = c_{\mathcal{R}}(n,m,\ell)$.*
(iii) *If $D = 2mn - \ell^2 > 0$ and $0 \leq \ell \leq 2m$, then $c_m^F(n,\ell) = c_{\mathcal{R}}(m,n,\ell)$.*

Proof. (i) For $j = \pm 1$ one of m_1, n_1 vanishes and the wall is $|v_2| = |n_1|\sigma_2$ with $|n_1| \geq \frac{1}{2}$, or $|v_2| = |m_1|\rho_2$ with $|m_1| \geq 1$, or $v_2 = 0$; none meets $\mathcal{R}$. For $|j| \geq 3$ the product $m_1 n_1$ is negative, so $|n_1\sigma_2 - m_1\rho_2| = |n_1|\sigma_2 + |m_1|\rho_2 \geq 2\sqrt{|m_1 n_1|\sigma_2\rho_2} = \sqrt{j^2-1}\,\sqrt{\sigma_2\rho_2}$, while on $\mathcal{R}$

$$|j\,v_2| < \tfrac{|j|}{2}\min(\rho_2, \tfrac{\sigma_2}{2}) \leq \tfrac{|j|}{2\sqrt{2}}\sqrt{\rho_2\sigma_2} < \sqrt{j^2-1}\,\sqrt{\rho_2\sigma_2}.$$

(ii) The substitution of Lemma 5.1 maps $\mathcal{R}$ to itself and exchanges the roles of m and n.

(iii) Let $\rho_2 \to \infty$ inside $\mathcal{R}$ with σ_2 and v_2 fixed. The integral over ρ_1 picks out φ_m, and $c_{\mathcal{R}}(m,n,\ell)$ is the coefficient of $x^n y^\ell$ of φ_m expanded in the strip $-\sigma_2 < v_2 < 0$, where $|p| <$

$|y|^{-1} < 1$. Write $\varphi_m = \varphi_m^P + \varphi_m^F$ with (8). The finite part is holomorphic in v, so its expansion does not depend on the strip. In the polar part, expand $\mathcal{A}_{2,m}$ in the strip: the term $s = 0$ is $y(1-y)^{-2} = \sum_{k\geq 1} k\, y^{-k}$, the terms $s \geq 1$ contain only powers $y^{2ms+1+k}$ with $k \geq 0$, and the terms $s \leq -1$, rewritten through $(1-p^s y)^{-2} = p^{-2s} y^{-2}(1-p^{-s}y^{-1})^{-2}$, only powers $y^{2ms-1-k}$ with $k \geq 0$. None of these exponents lies in $[0, 2m]$, and the factor $1/g(\sigma/2)$ does not change the power of y. Hence the polar part contributes nothing at y^ℓ with $0 \leq \ell \leq 2m$. □

5.3. **Least representatives.**

Lemma 5.4. *Let (m, n, ℓ) be a charge with $m \geq 1$, $|\ell| \leq m$, $|\ell| \leq n$ and $D = 2mn - \ell^2 > 0$. Then it lies in the set $\mathcal{A}$ of* (10).

Proof. For $m_1 = 0$ the bracket in (10) is $m^2(2s+1)^2 \geq m^2$. Let $m_1 \neq 0$, so $|m_1| \geq 2$, and let d be the distance from $m_1\ell$ to the nearest odd multiple of m; the bracket is at least $d^2 + m_1^2 D$. If $|m_1\ell| \leq m$, then $d \geq m - |m_1\ell|$ and

$$d^2 + m_1^2 D \;\geq\; m^2 - 2m|m_1\ell| + m_1^2\ell^2 + m_1^2(2mn - \ell^2) = m^2 + 2m|m_1|\big(|m_1|n - |\ell|\big) \;\geq\; m^2.$$

If $|m_1\ell| > m$, then $D \geq 2m|\ell| - \ell^2 \geq m|\ell|$ and $m_1^2 D \geq m|m_1| \cdot |m_1\ell| > 2m^2$. □

Definition 5.5. A charge (m, n, ℓ) is *reduced* if $1 \leq m \leq n$, $0 \leq \ell \leq m$ and $D = 2mn - \ell^2 > 0$. For a reduced charge put $x = \sqrt{D}/m$ and $\theta = \pi\ell/m$. Since $D \geq 2m^2 - m^2$, every reduced charge has $x \geq 1$.

Proposition 5.6. *For every charge (Q, P) with $D > 0$ there is a reduced charge (m', n', ℓ') with the same discriminant and $\ell' \equiv Q \cdot P \pmod 2$ such that*

$$d^*(Q, P) = (-1)^{\ell'+1} c_{m'}^F(n', \ell') = (-1)^{\ell'+1} c_{\mathcal{R}}(m', n', \ell').$$

Proof. In the orbit of (Q, P) choose a charge with least $m = Q^2$; since $D > 0$ the form $f(a, b) = (aQ + bP)^2 = a^2 m + 2ab\ell + 2b^2 n$ is positive definite, so the least value exists. The matrices $\left(\begin{smallmatrix} 1 & 0 \\ c & 1 \end{smallmatrix}\right)$ with c even fix Q and send ℓ to $\ell + cm$; so we may assume $|\ell| \leq m$. The matrices $\left(\begin{smallmatrix} 1 & \pm 1 \\ 0 & 1 \end{smallmatrix}\right)$ send m to $m \pm 2\ell + 2n$, and minimality gives $n \geq |\ell|$. By Lemma 5.4 the charge lies in $\mathcal{A}$, so $d^* = (-1)^{\ell+1} c_m^F(n, \ell)$ by (11). The finite part is a mock Jacobi form, even in v, so $c_m^F(n, \ell) = c_m^F(n, |\ell|)$. If $m \leq n$, the charge $(m, n, |\ell|)$ is reduced. If $n < m$, then $0 \leq |\ell| \leq n$ and Lemma 5.3 gives $c_m^F(n, |\ell|) = c_{\mathcal{R}}(m, n, |\ell|) = c_{\mathcal{R}}(n, m, |\ell|) = c_n^F(m, |\ell|)$, and $(n, m, |\ell|)$ is reduced. The shifts change ℓ by even multiples of m and the sign change does not alter the parity, so $\ell' \equiv Q \cdot P \pmod 2$. □

So for the rest of the paper we write $d^*(m, n, \ell) = (-1)^{\ell+1} c_m^F(n, \ell)$ for a reduced charge, and the conjecture is the statement $d^*(m, n, \ell) > 0$ on reduced charges.

5.4. **Bounded discriminant.**

Proof of Theorem B. A reduced charge with $D \leq 800$ has $m \leq \sqrt{D} \leq \lfloor\sqrt{800}\rfloor$ and $n = (D + \ell^2)/2m \leq (D + m^2)/2m$, so there are finitely many. For each $\Gamma_1(2)$ orbit with $D \leq 800$ the program of Appendix A finds a charge of least Q^2 satisfying (10) with $0 \leq \ell < 2m$, and computes $c_{\mathcal{R}}(m, n, \ell)$ exactly from the product formula, in the cheaper of the two orientations allowed by Lemma 5.3. For $D \leq 800$ there are 22972 orbits; in every one of them $(-1)^{\ell+1} c_{\mathcal{R}}(m, n, \ell) > 0$. By Lemma 5.3(iii) and (11) these numbers are the single-centred indices.[7] □

[7]Every orbit is found with all its representatives inside the computed box, and the program checks that they give the same value; at $Q^2 = 1, 2$ it reproduces the closed formulas of Propositions 3.1 and 4.2.

6. Farey walls

The polar coefficients that enter the Rademacher series are sums over matrices of $SL_2(\mathbb{Z})$ attached to Farey intervals, the continued-fraction walls of [9, 11]. This section collects the elementary facts about them that the estimates need.

6.1. **The walls of a polar term.** Fix $m \geq 1$, $\nu \in \frac{1}{2}\mathbb{Z}$ and $\tilde{\ell} \in \mathbb{Z}$ with $0 \leq \tilde{\ell} < 2m$ and

$$\tilde{D} = 4m\nu - \tilde{\ell}^2 < 0, \qquad t = -\tilde{D}, \qquad \omega_0 = \frac{\tilde{\ell}}{2m}, \qquad \delta = \frac{\sqrt{t}}{2m}.$$

Let $\mathcal{W}(m, \nu, \tilde{\ell})$ be the set of $G = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$ with $r, s \geq 1$ and $q/s \leq \omega_0 < p/r$; for each such G put

$$m_G = r^2\nu + p^2 m - pr\tilde{\ell}, \qquad n_G = s^2\nu + q^2 m - qs\tilde{\ell}, \qquad \ell_G = -2rs\nu - 2pqm + (ps+qr)\tilde{\ell}. \tag{28}$$

These are the coefficients of the quadratic form $\phi(X, Y) = \nu X^2 - \tilde{\ell}XY + mY^2$ transformed by G: $m_G = \phi(r, p)$, $n_G = \phi(s, q)$, and ℓ_G is the cross term up to sign.

Lemma 6.1. *Let $G \in \mathcal{W}(m, \nu, \tilde{\ell})$ and $U = p/r - \omega_0$, $W = \omega_0 - q/s$, so that $U > 0$, $W \geq 0$ and $U + W = 1/rs$.*

(i) $\ell_G^2 - 4m_G n_G = t$.

(ii) $m_G = mr^2(U^2 - \delta^2)$, $n_G = ms^2(W^2 - \delta^2)$, $\ell_G = 2mrs(UW + \delta^2) > 0$, *and* $m = m_G s^2 + n_G r^2 + \ell_G rs$.

(iii) $E_G := m - m_G - n_G - \ell_G = m\big[1 - (rU + sW)^2 + \delta^2(r - s)^2\big] \geq 0$.

(iv) *If $m_G, n_G \geq -1$, put $a = m_G$, $b = n_G$, $S = a + b + 2$ and $t_0 = m^2 + 4m$. Then*

$$t_0 - t = S(2m - S) + E_G^2 + 2E_G\ell_G + 4(a+1)(b+1), \tag{29}$$

so $t \leq t_0$; moreover $t_0 - t \leq 2mS$ when $r = s = 1$.

(v) *If $m_G, n_G \geq -1$, then $a \leq m/s^2$, $b \leq m/r^2$ and $r, s \leq 2m$.*

(vi) *If $m_G, n_G \geq -1$ and $rs \geq 2$, then $\delta \leq \frac{1}{4} + \frac{5}{2m}$, that is $\sqrt{t} \leq \frac{m}{2} + 5$.*

(vii) *If $r = s = 1$, then $G = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, $a = \nu + m - \tilde{\ell}$, $b = \nu$, $\ell_G = m - a - b$, $E_G = 0$ and $\tilde{\ell} = m + b - a$. Conversely every pair (a, b) of integers with $a, b \geq -1$, $k := m - a - b \geq 1$, $t = k^2 - 4ab > 0$ and $-m \leq b - a < m$ arises from exactly one polar term with $\nu = b \in \mathbb{Z}$ in this way.*

Proof. (i) $\phi \circ G$ has the same discriminant as ϕ. (ii) $\phi(r, p) = r^2\big(\nu - \tilde{\ell}u + mu^2\big)$ with $u = p/r$, and $\nu - \tilde{\ell}u + mu^2 = m\big((u - \omega_0)^2 - \delta^2\big)$ because $\tilde{\ell}^2 - 4m\nu = 4m^2\delta^2$; similarly for n_G. For ℓ_G write $\ell_G = 2mrs\big[(p/r - \omega_0)(\omega_0 - q/s) + \delta^2\big]$, which is (28) after expanding and using $ps - qr = 1$. The last identity follows from $\phi(r, p)s^2 + \phi(s, q)r^2 + \ell_G rs = m(ps - qr)^2$. (iii) Insert (ii) in the definition and use $U + W = 1/rs$; then $rU + sW \leq \max(r, s)(U + W) = 1/\min(r, s) \leq 1$. (iv) By (i), $t = \ell_G^2 - 4ab$ with $\ell_G = m - a - b - E_G = m + 2 - S - E_G$; expand $t_0 - t = m^2 + 4m - (m + 2 - S - E_G)^2 + 4ab$ and use $4ab = 4(a+1)(b+1) - 4S + 4$. For $r = s = 1$, $E_G = 0$ and $4(a+1)(b+1) \leq S^2$. (v) $a = mr^2(U^2 - \delta^2) \leq mr^2U^2 \leq m/s^2$, and $b \leq m/r^2$ in the same way. Since $t \geq 1$, $\delta \geq 1/2m$. From $a \geq -1$, $\delta^2 \leq U^2 + 1/(mr^2)$, and from $b \geq -1$, $\delta^2 \leq W^2 + 1/(ms^2)$. One of U, W is at most $1/(2rs)$; if it is U, the first inequality gives $r^2 \leq 4m^2(\frac{1}{4s^2} + \frac{1}{m}) \leq m^2 + 4m$, so $r \leq m + 1$, and the second with $W \leq 1/rs$ gives $s^2 \leq 4m^2(1 + \frac{1}{m})$, so $s \leq 2m$. The other case is symmetric. (vi) If $\delta \leq \frac{1}{4}$ there is nothing to prove. Otherwise, from $a \geq -1$, $\delta - U \leq 1/(mr^2(\delta + U)) \leq 4/(mr^2)$, and from $b \geq -1$, $\delta - W \leq 4/(ms^2)$. Adding, $2\delta \leq \frac{1}{rs} + \frac{4}{m}\big(\frac{1}{r^2} + \frac{1}{s^2}\big) \leq \frac{1}{2} + \frac{5}{m}$ when $rs \geq 2$. (vii) $p - q = 1$ and $q \leq \omega_0 < q + 1$ with $0 \leq \omega_0 < 1$ force $q = 0$, $p = 1$. The formulas follow from (28). Given (a, b), put $\nu = b$ and $\tilde{\ell} = m + b - a \in [0, 2m)$; then $\tilde{D} = 4mb - (m + b - a)^2 = -(k^2 - 4ab)$. $\square$

Remark 6.2. Lemma 6.1(iii) is the reason why the leading family dominates: by (29) the walls with $r = s = 1$ are exactly those for which $t_0 - t$ is as small as the pair (a, b) allows, and by (vi) every other wall has $\sqrt{t}$ at most about half of $\sqrt{t_0}$. Figure 1 shows the geometry.

6.2. **Polar coefficients.** Let $d_1(k)$, $k \geq -1$, be the coefficients of

$$\frac{1}{\eta(\tau)^8\eta(2\tau)^8} = \sum_{k\geq -1} d_1(k)\, q^k = q^{-1} + 8 + 52q + 256q^2 + 1122q^3 + \cdots, \tag{30}$$

so that $d_1(k) = \hat{f}(k)$ in the notation of (8), and put $d_1(k) = 0$ unless $k \geq -1$ is an integer. For a polar term $(\nu, \tilde{\ell})$ define

$$a_S(\nu, \tilde{\ell}) = \frac{1}{16} \sum_{G\in\mathcal{W}(m,\nu,\tilde{\ell})} \ell_G\, d_1(m_G)\, d_1(n_G) \qquad (\nu \in \mathbb{Z}), \tag{31}$$

$$a_I(\nu, \tilde{\ell}) = \sum_{G\in\mathcal{W}(m,\nu,\tilde{\ell})} \ell_G \Big([r \text{ even}]\, d_1(m_G)\, d_1(2n_G) + [s \text{ even}]\, d_1(2m_G)\, d_1(n_G) \Big) \qquad (\nu \in \tfrac{1}{2}\mathbb{Z}), \tag{32}$$

and $a_T(\nu, \tilde{\ell}) = (-1)^{2\nu} a_I(\nu, \tilde{\ell})$. The sums are finite by Lemma 6.1(v). Formula (31) is the continued-fraction formula of [9, (2.43)] for $1/\Phi_{10}$ with the coefficients of $1/\eta^{24}$ replaced by d_1 and with no congruence condition on G, and (32) is its parity-restricted form [9, (4.21)].[8]

7. The level-two Rademacher expansion

This section states the one input of Theorem C that is not proved here, explains where it comes from, and records the evidence for it.

7.1. **Frames and Weil sums.** Let $\Gamma^0(2)$ be the group of matrices of $SL_2(\mathbb{Z})$ with even upper right entry. It has index three, and

$$R_I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad R_T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad R_S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

represent its right cosets. For $M \in SL_2(\mathbb{Z})$ the *frame* of M is the $F \in \{I, T, S\}$ with $MR_F^{-1} \in \Gamma^0(2)$. The three frames correspond to the expansions of a form for $\Gamma^0(2)$ at the cusp ∞ (frames I and T, which differ by a half period) and at the cusp 0 (frame S).[9]

For $\gamma \geq 1$ let $\mathcal{C}_\gamma$ be the set of pairs (γ, δ) with $-\delta \in \{0, 1, \ldots, 2\gamma - 1\}$ and $\gcd(\gamma, \delta) = 1$. For each such pair let $\alpha \in \{0, \ldots, \gamma - 1\}$ be the inverse of δ modulo γ (with $\alpha = 0$ if $\gamma = 1$), let $\beta = (\alpha\delta - 1)/\gamma$, and let $F(\gamma, \delta)$ be the frame of $\left(\begin{smallmatrix} \delta & -\beta \\ -\gamma & \alpha \end{smallmatrix}\right)$. For $\tilde{\ell}, \ell \in \mathbb{Z}$ put

$$W_{\gamma,\delta}(\tilde{\ell}, \ell) = \frac{1}{\sqrt{2m\gamma i}} \sum_{T=0}^{\gamma-1} \mathbf{e}\Big(\frac{\alpha x^2 - 2\ell x + \delta\ell^2}{4m\gamma} \Big), \qquad x = \tilde{\ell} - 2mT, \tag{33}$$

with the principal square root. This is the matrix coefficient of the Weil representation attached to the lattice $\sqrt{2m}\,\mathbb{Z}$, evaluated on the inverse of $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)$ [9, (2.41)]; it depends on $\tilde{\ell}$ only modulo $2m$.

[8] In (32) the condition that r or s be even selects the walls that are images of the basic wall under $\Gamma_0(2)$ or its conjugate; the factor $d_1(2n_G)$ vanishes unless $2n_G$ is an integer.

[9] The duality group $\Gamma_1(2)$ acts on the charges; the finite part $\varphi_m^F(\sigma, v)$ is a function of σ on which the relevant group, after the substitution $x = \mathbf{e}(\sigma/2)$, is $\Gamma^0(2)$. The cusp ∞ has width two, which is why it carries two frames.

Lemma 7.1. *$|W_{1,\delta}| = (2m)^{-1/2}$, and $|W_{\gamma,\delta}(\tilde\ell,\ell)| \le \sqrt{\gcd(m,\gamma)/m}$ for every $\gamma \ge 1$.*

Proof. For $\gamma = 1$ the sum has one term. In general, $\alpha x^2/(4m\gamma) = \alpha\tilde\ell^2/(4m\gamma) - \alpha\tilde\ell T/\gamma + \alpha m T^2/\gamma$, so up to a factor of modulus one the sum is $\sum_{T \bmod \gamma} \mathbf{e}\big((\alpha m T^2 + bT)/\gamma\big)$ for an integer b. A quadratic Gauss sum modulo γ with leading coefficient αm, $\gcd(\alpha,\gamma) = 1$, has modulus at most $\sqrt{2\gcd(m,\gamma)\gamma}$ [1, Chapter 1]. Dividing by $\sqrt{2m\gamma}$ gives the claim. □

7.2. **Polar data and shadows.** For $m \ge 1$ the polar terms in each frame are the pairs $(\tilde\nu,\tilde\ell)$ with $0 \le \tilde\ell < 2m$ and $\tilde D = 4m\tilde\nu - \tilde\ell^2 < 0$, where $\tilde\nu \in \mathbb{Z}$ in frame S and $\tilde\nu \in \frac12\mathbb{Z}$ in frames I and T. Their coefficients are the wall sums of Section 6:

$$a_S(\tilde\nu,\tilde\ell) \text{ from (31)}, \qquad a_I(\tilde\nu,\tilde\ell) \text{ from (32)}, \qquad a_T(\tilde\nu,\tilde\ell) = (-1)^{2\tilde\nu} a_I(\tilde\nu,\tilde\ell). \tag{34}$$

By Lemma 6.1(iv) a wall with $m_G, n_G \ge -1$ has $t \le t_0 = m^2 + 4m$, so only finitely many polar terms have a nonzero coefficient.

The shadow of the finite part is governed by the coefficient $\hat f(m) = d_1(m)$ of the polar part (8). In the three frames it enters with the constants and the orders

$$(c_S,\mu_S) = \big(\tfrac{1}{16}\hat f(m),\, 1\big), \qquad (c_I,\mu_I) = \big(\hat f(m),\, \tfrac12\big), \qquad (c_T,\mu_T) = \big(-\hat f(m),\, \tfrac12\big). \tag{35}$$

The number μ_F is the order of the pole of $1/g(\sigma/2)$ in the local parameter of the frame: $1/g$ has the term $x^{-1} = \mathbf{e}(-\sigma/2)$ at ∞, and its transform under $\sigma \mapsto -1/\sigma$ starts with $\mathbf{e}(-\sigma)/16$, by $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$.[10] For $\gamma \ge 1$, $(\gamma,\delta) \in \mathcal{C}_\gamma$ and $g \in \mathbb{Z}$ put

$$f_{\gamma,g}(u) = \begin{cases} \dfrac{\pi^2}{\sinh^2\big(\frac{\pi u}{\gamma} - \frac{i\pi g}{2m\gamma}\big)}, & 2m\gamma \nmid g, \\[2ex] \dfrac{\pi^2}{\sinh^2(\pi u/\gamma)} - \dfrac{\gamma^2}{u^2}, & 2m\gamma \mid g. \end{cases}$$

7.3. **The hypothesis.** For a charge (m,n,ℓ) with $m \ge 1$, $0 \le \ell \le m$ and $D = 2mn - \ell^2 > 0$ define

$$\mathcal{M}(m,n,\ell) = \sum_{\gamma\ge1} \sum_{(\gamma,\delta)\in\mathcal{C}_\gamma} \sum_{(\tilde\nu,\tilde\ell)} a_F(\tilde\nu,\tilde\ell)\, \mathbf{e}\Big(\frac{\alpha\tilde D + \delta D}{4m\gamma}\Big) W_{\gamma,\delta}(\tilde\ell,\ell)\, \frac{2\pi}{\gamma}\Big(\frac{|\tilde D|}{D}\Big)^{\frac{15}{4}} I_{\frac{15}{2}}\Big(\frac{\pi\sqrt{|\tilde D|D}}{m\gamma}\Big), \tag{36}$$

$$\mathcal{K}_2(m,n,\ell) = -\sqrt{2m}\sum_{\gamma\ge1} \gamma^{-\frac12} \sum_{(\gamma,\delta)\in\mathcal{C}_\gamma} c_F\, \mathbf{e}\Big(\frac{-4m\mu_F\alpha + \delta D}{4m\gamma}\Big) W_{\gamma,\delta}(0,\ell)\Big(\frac{4m\mu_F}{D}\Big)^4 I_8\Big(\frac{2\pi}{\gamma}\sqrt{\frac{\mu_F D}{m}}\Big), \tag{37}$$

$$\mathcal{K}_3(m,n,\ell) = \frac{1}{2\pi}\sum_{\gamma\ge1} \gamma^{-2} \sum_{(\gamma,\delta)\in\mathcal{C}_\gamma} \sum_{g=-m\gamma}^{m\gamma-1} c_F\, \mathbf{e}\Big(\frac{-\alpha(4m\mu_F + g^2) + \delta D}{4m\gamma}\Big) W_{\gamma,\delta}(g,\ell)\Big(\frac{4m\mu_F}{D}\Big)^{\frac{17}{4}} \times \int_{-\sqrt{\mu_F/m}}^{\sqrt{\mu_F/m}} f_{\gamma,g}(u)\Big(1 - \frac{mu^2}{\mu_F}\Big)^{\frac{17}{4}} I_{\frac{17}{2}}\Big(\frac{2\pi}{\gamma}\sqrt{\frac{(\mu_F - mu^2)D}{m}}\Big)\, du, \tag{38}$$

where in each line $F = F(\gamma,\delta)$ and α,β are attached to (γ,δ) as above.

[10] With $\tau = \sigma/2$, the substitution $\sigma \mapsto -1/\sigma$ is $\tau \mapsto -1/4\tau$, and $g(-1/4\tau) = \eta(-1/4\tau)^8\eta(-1/2\tau)^8 = 2^{12}\tau^8\eta(\sigma)^8\eta(2\sigma)^8$, whose expansion starts with $\mathbf{e}(\sigma)$.

Hypothesis R. For every charge (m,n,ℓ) with $m\geq 1$, $0\leq\ell\leq m$ and $D=2mn-\ell^2>0$,

$$(-1)^{\ell+1}c_m^F(n,\ell)=\tfrac{1}{2}(-1)^{\ell}\sqrt{i}\,\big(\mathcal{M}+\mathcal{K}_2+\mathcal{K}_3\big)(m,n,\ell). \tag{39}$$

Lemma 7.2. *The three series* (36)–(38) *converge absolutely.*

Proof. There are at most 2γ pairs in $\mathcal{C}_\gamma$, the polar terms are finitely many, and $|W|\leq 1$ by Lemma 7.1. Since $I_\nu(z)/z^\nu$ is increasing, $I_\nu(z/\gamma)\leq\gamma^{-\nu}I_\nu(z)$ for $\gamma\geq 1$. So the γ-th term of (36) is $O(\gamma\cdot\gamma^{-1}\cdot\gamma^{-15/2})$, that of (37) is $O(\gamma^{1-1/2-8})$. In (38), for $2m\gamma\nmid g$ one has $|\sinh(a-ib)|^2=\sinh^2 a+\sin^2 b\geq\sin^2 b$, so $|f_{\gamma,g}|\leq\pi^2\csc^2(\pi g/2m\gamma)$, and for $g=0$ the function $\pi^2/\sinh^2 y-\pi^2/y^2$ takes values in $[-\pi^2/3,0)$. With the identity $\sum_{g=1}^{N-1}\csc^2(\pi g/N)=(N^2-1)/3$ for $N=2m\gamma$,

$$\sum_{g=-m\gamma}^{m\gamma-1}\sup_u|f_{\gamma,g}(u)|\leq\frac{\pi^2(4m^2\gamma^2-1)}{3}+\frac{\pi^2}{3}=\frac{4\pi^2m^2\gamma^2}{3}, \tag{40}$$

so the γ-th term of (38) is $O(\gamma\cdot\gamma^{-2}\cdot\gamma^{2}\cdot\gamma^{-17/2})$. Only absolute bounds are used below, except for the terms with $\gamma=1$ in frame S, which are computed explicitly in Section 8. □

Remark 7.3. Hypothesis R has three ingredients.

- (H) The shape of the expansion: a mixed mock Jacobi form of weight -6 and index m for $\Gamma^0(2)$, whose shadow is a multiple of the vector of unary theta functions of index m with coefficient $1/g$, has Fourier coefficients given by the vector-valued Rademacher series over the three frames, with the main Bessel terms $I_{15/2}$ from the principal parts and the correction terms I_8 and $I_{17/2}$ from the shadow. For a weakly holomorphic Jacobi form at level N this is the classical circle method of Rademacher and Zuckerman [25, 36], written for Jacobi forms by its theta decomposition. For mixed mock Jacobi forms on $SL_2(\mathbb{Z})$, which is the case of $1/\Phi_{10}$, the expansion was proved by Bringmann and Manschot [5] and used by Ferrari and Reys [18]; the correction terms come from the non-holomorphic completion of Zwegers [37]. At level two it is stated in [9, §4], where it is assumed in the derivation of the immortal index, and we do not know a proof.
- (P0) The principal part at the cusp 0 is given by the wall sum (31): the continued-fraction formula of [9, (2.43)] with d_1 in place of the coefficients of $1/\eta^{24}$.
- (P∞) The principal part at the cusp ∞ is given by the parity-restricted wall sum (32).

The constants (35) are part of (H): they are the shadow coefficients of φ_m^F in the three frames, read off from the polar part (8), whose shadow is the image of $\hat{f}(m)\mathcal{A}_{2,m}/g$ under the completion of the Appell–Lerch sum [37, 13].

7.4. **Evidence.** We have tested the three ingredients separately and together.

- (a) For $m=1,2$ the coefficients $c_m^F(n,\ell)$ are known in closed form (Propositions 3.1 and 4.2). The right side of (39), truncated at $\gamma\leq 8$ in $\mathcal{M}$ and at $\gamma\leq 4$ in $\mathcal{K}_2$ and $\mathcal{K}_3$, agrees with them to relative precision 10^{-21} for every $n\leq 42$.
- (b) For $m=3$ ($n=8,12$) and $m=4$ ($n=10,12$), with the same truncation ($\gamma\leq 8$ in $\mathcal{M}$, $\gamma\leq 4$ in $\mathcal{K}_2,\mathcal{K}_3$) and with $c_m^F(n,\ell)$ computed exactly from the product formula, the relative difference is below 10^{-13} for every $0\leq\ell\leq m$. The size of the omitted terms, estimated by Lemma 7.2, is of the same order, so the agreement is as good as the truncation allows.
- (c) The polar coefficients at the cusp ∞ are Fourier coefficients of φ_m^F itself, computable from the product formula. On all 373 polar terms with $m\leq 14$ they agree with (32).

(d) The polar coefficients at the cusp 0 can be computed by expanding φ_m at that cusp, where the twisted genera (2) are permuted by $\sigma \mapsto -1/\sigma$, and subtracting the transform of the polar part. On all 168 polar terms with $m \le 10$ they agree with (31).

All of these are exact or high-precision computations; the programs are in the archive [3].

Remark 7.4. Without (P0) and (P∞), (H) alone would still give an expansion whose principal parts are *some* finite sets of numbers. What the proof of Theorem C uses from (P0) is exactly the structure of Lemma 6.1: the leading family of walls and the bound (vi) for the others. From (P∞) it uses only that every wall in it has r or s even, hence $rs \ge 2$, so that Lemma 6.1(vi) applies.

8. The leading family and an eta quotient

Fix $m \ge 1$ and a charge (m, n, ℓ) with $0 \le \ell \le m$ and $D > 0$; write $x = \sqrt{D}/m$, $\theta = \pi\ell/m$ and $t_0 = m^2 + 4m$. The *extreme term* is

$$\mathcal{E}(D,m) = \frac{\pi}{\sqrt{2m}} \cdot \frac{m+2}{16} \Big(\frac{t_0}{D}\Big)^{\frac{15}{4}} I_{\frac{15}{2}}\big(\pi x \sqrt{t_0}\big) > 0. \tag{41}$$

It is the contribution to $\frac{1}{2}(-1)^\ell\sqrt{i}\,\mathcal{M}$ of the frame S, $\gamma = 1$, and the most polar term $\tilde\nu = -1$, $\tilde\ell = m$, whose only wall is $G_0 = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ with $m_{G_0} = n_{G_0} = -1$ and $\ell_{G_0} = m + 2$.

8.1. **The terms with $\gamma = 1$.** For $\gamma = 1$ the set $\mathcal{C}_1$ consists of $\delta = 0$ and $\delta = -1$, both with $\alpha = 0$ and $\beta = -1$. The matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ has frame S, and $\left(\begin{smallmatrix} -1 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ has frame T. So the terms with $\gamma = 1$ are one term in frame S and one in frame T.

In frame S, $W_{1,0}(\tilde\ell, \ell) = (2mi)^{-1/2}\mathbf{e}(-\ell\tilde\ell/2m)$ and the phase in (36) is 1. Since $\sqrt{i}/\sqrt{2mi} = 1/\sqrt{2m}$,

$$\tfrac{1}{2}(-1)^\ell\sqrt{i}\,\mathcal{M}_{\gamma=1,S} = \frac{\pi}{\sqrt{2m}} \sum_{(\tilde\nu,\tilde\ell)} a_S(\tilde\nu,\tilde\ell)\,(-1)^\ell \mathbf{e}\Big(-\frac{\ell\tilde\ell}{2m}\Big)\Big(\frac{t}{D}\Big)^{\frac{15}{4}} I_{\frac{15}{2}}(\pi x\sqrt{t}), \qquad t = -\tilde D. \tag{42}$$

Split $a_S = a_S^{\mathrm{lead}} + a_S^{\mathrm{rest}}$, where a_S^{lead} is the term of G_0 in (31) and a_S^{rest} the sum over the walls with $rs \ge 2$. Lemma 6.1(vii) identifies the polar terms carrying G_0 with the set

$$\mathcal{L}_m = \big\{(a,b) \in \mathbb{Z}^2 :\ a, b \ge -1,\ k = m - a - b \ge 1,\ t_{a,b} = k^2 - 4ab > 0,\ -m \le b - a < m\big\},$$

by $\tilde\nu = b$, $\tilde\ell = m + b - a$, with $a_S^{\mathrm{lead}} = \frac{1}{16} k\, d_1(a) d_1(b)$ and $t = t_{a,b}$. For $(a,b) \in \mathcal{L}_m$ put $S = a + b + 2$ and

$$\rho_{a,b}(x) = \frac{k}{m+2}\Big(\frac{t_{a,b}}{t_0}\Big)^{\frac{15}{4}} \frac{I_{15/2}(\pi x\sqrt{t_{a,b}})}{I_{15/2}(\pi x\sqrt{t_0})}. \tag{43}$$

Theorem 8.1. *The leading part of* (42) *is*

$$\begin{aligned} &\frac{\pi}{\sqrt{2m}} \sum_{(\tilde\nu,\tilde\ell)} a_S^{\mathrm{lead}}(\tilde\nu,\tilde\ell)\,(-1)^\ell \mathbf{e}\Big(-\frac{\ell\tilde\ell}{2m}\Big)\Big(\frac{t}{D}\Big)^{\frac{15}{4}} I_{\frac{15}{2}}(\pi x\sqrt{t}) \\ &\qquad = \mathcal{E}(D,m) \sum_{(a,b)\in\mathcal{L}_m} d_1(a) d_1(b)\, \rho_{a,b}(x) \cos\big((a-b)\theta\big). \end{aligned} \tag{44}$$

Moreover $\rho_{-1,-1} = 1$, and with $z = e^{-\pi x + i\theta}$,

$$\sum_{a,b\ge -1} d_1(a) d_1(b)\, e^{-\pi x(a+b+2)} \cos\big((a-b)\theta\big) = |P(z)|^2, \qquad P(z) = \prod_{n\ge 1} (1-z^n)^{-8}(1-z^{2n})^{-8}. \tag{45}$$

Proof. With $\tilde{\ell} = m + b - a$ one has $(-1)^{\ell}\mathbf{e}(-\ell\tilde{\ell}/2m) = (-1)^{\ell}(-1)^{\ell}\mathbf{e}(\ell(a-b)/2m) = e^{i(a-b)\theta}$. Dividing the term of (a,b) by $\mathcal{E}(D,m)$ leaves $d_1(a)d_1(b)\rho_{a,b}(x)e^{i(a-b)\theta}$. The conditions defining $\mathcal{L}_m$ other than $-m \le b-a < m$ are symmetric in (a,b), and the weight $d_1(a)d_1(b)\rho_{a,b}$ is symmetric; so the terms of (a,b) and (b,a) are complex conjugate when both lie in $\mathcal{L}_m$, and the only unpaired terms have $|a-b| = m$, where $e^{i(a-b)\theta} = e^{\pm i\pi\ell}$ is real. Hence the sum equals its real part, which is (44). For $a = b = -1$ one has $k = m+2$ and $t = (m+2)^2 - 4 = t_0$. Finally $P(z) = \sum_{a\ge -1} d_1(a) z^{a+1}$ by (30), so $|P(z)|^2 = \sum_{a,b} d_1(a)d_1(b) z^{a+1}\bar{z}^{b+1}$; since $z^{a+1}\bar{z}^{b+1} = e^{-\pi x S}e^{i(a-b)\theta}$ and the sum is symmetric, it is real and equals (45). □

The comparison between (44) and (45) rests on the following estimates for the exponents. Write $\Delta_s = \sqrt{t_0} - \sqrt{t_{a,b}}$.

Lemma 8.2. *Let $(a,b) \in \mathcal{L}_m$ and $S = a+b+2 \ge 1$. Then*

(i) $t_0 - t_{a,b} = S(2m-S) + 4(a+1)(b+1) \le 2mS$, *and* $t_{a,b} \ge m(m+4-2S) = t_0 - 2mS$;

(ii) $S - \Delta_s \le \eta_- := \dfrac{S(S+4)}{2m+4}$;

(iii) $\Delta_s - S \le \eta_+$, *where* $\eta_+ = \max\big(0, \sqrt{m(m+4)} - \sqrt{m(m+4-2S)} - S\big)$ *if* $2S \le m+4$ *and* $\eta_+ = m+2-S$ *otherwise;*

(iv) $\Delta_s \ge \dfrac{S(2m-S)}{2(m+2)}$, *and* $S \le m+1$.

Proof. (i) is Lemma 6.1(iv) with $E_{G_0} = 0$, together with $4(a+1)(b+1) \le S^2$; and $4ab \le (a+b)^2 = (S-2)^2$ gives $t_{a,b} = (m+2-S)^2 - 4ab \ge (m+2-S)^2 - (S-2)^2 = m(m+4-2S)$. (iv) $S \le m+1$ because $k \ge 1$; then $\Delta_s = (t_0 - t)/(\sqrt{t_0} + \sqrt{t}) \ge S(2m-S)/(2\sqrt{t_0})$ and $\sqrt{t_0} \le m+2$. (ii) follows from (iv): $S - \Delta_s \le S - \frac{S(2m-S)}{2m+4} = \frac{S(S+4)}{2m+4}$. (iii) If $2S \le m+4$, then $\sqrt{t} \ge \sqrt{m(m+4-2S)}$ by (i); otherwise $\Delta_s \le \sqrt{t_0} \le m+2$. □

Proposition 8.3. *On the closed disc $|z| \le e^{-\pi}$,*

$$|P(z)|^2 \;\ge\; 0.53816,$$

and the minimum is attained at $z = -e^{-\pi}$, where $|P|^2 = 0.53887\ldots$.

Proof. P is holomorphic and zero-free on $|z| < 1$, so by the minimum modulus principle the minimum of $|P|$ on the disc is attained on the circle $|z| = r_0 = e^{-\pi}$. Since the coefficients of P and zP' are nonnegative, $|P(z)| \le P(r_0)$ and $|zP'(z)| \le r_0 P'(r_0)$ on the circle, so $\theta \mapsto |P(r_0 e^{i\theta})|^2$ is Lipschitz with constant $L = 2P(r_0)\, r_0 P'(r_0)$. Since $|P(\bar{z})| = |P(z)|$ it suffices to take $\theta \in [0,\pi]$. We evaluate P at $\theta_k = \pi k/4000$, $0 \le k \le 4000$, in ball arithmetic [21], with the product truncated at $n = 60$ and the tail enclosed by $|\log \prod_{n>60}(1-z^n)^{-8}(1-z^{2n})^{-8}| \le 16 r_0^{61}/((1-r_0)(1-r_0^{61}))$. The least lower bound obtained, minus $L\pi/8000$, is 0.53816. □

Remark 8.4. The function $1/(\eta(\tau)^8\eta(2\tau)^8)$, whose coefficients are d_1, is the generating function of the electric half-BPS degeneracies of the model [28]. Theorem 8.1 therefore says that, to leading order, the single-centred index of a quarter-BPS black hole is the extreme Bessel term times the squared modulus of the half-BPS generating function, evaluated at the point $z = e^{-\pi\sqrt{D}/Q^2 + i\pi Q\cdot P/Q^2}$. The positivity of $|P|^2$ is automatic; what has to be proved is that the corrections cannot overturn it, and Proposition 8.3 gives the room.

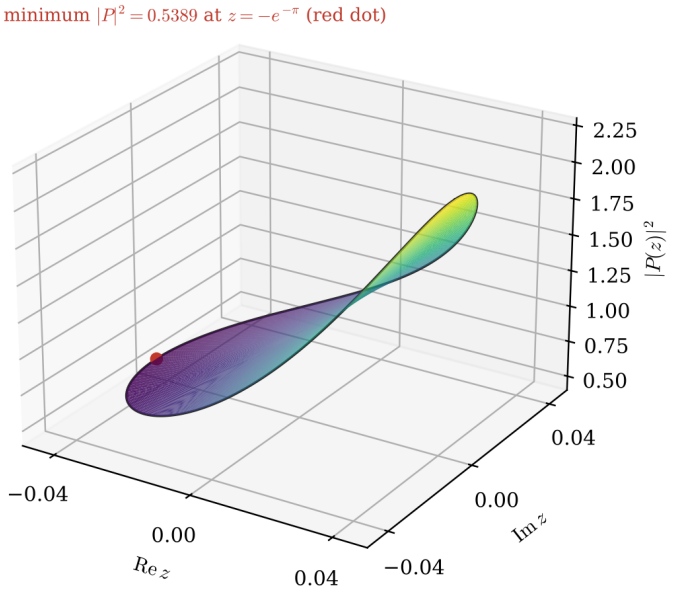


FIGURE 3. The surface $|P(z)|^2$ over the disc $|z| \le e^{-\pi}$, where $z = e^{-\pi x + i\theta}$ with $x \ge 1$. The rim is the image of $x = 1$; the lowest point is $z = -e^{-\pi}$, which corresponds to $x = 1$ and $\ell = m$ odd.

9. Facts on Bessel functions

All Bessel functions below have real order $\nu \ge \frac{1}{2}$ and positive argument. We use

$$g(y) = \sqrt{2\pi y}\, e^{-y} I_{15/2}(y), \qquad \phi(y) = \frac{y\, g'(y)}{g(y)}. \tag{46}$$

Lemma 9.1. (i) *$I_\nu(z)/z^\nu$ is increasing; hence $I_\nu(z/\gamma) \le \gamma^{-\nu} I_\nu(z)$ for $\gamma \ge 1$.*

(ii) *$z\, I_\nu'(z)/I_\nu(z)$ is increasing in z. Consequently, for $0 < c \le 1$ the function $y \mapsto I_\nu(cy)/I_\nu(y)$ is nonincreasing.*

(iii) *For $\nu' \ge \nu$, $z\, I_{\nu'}'(z)/I_{\nu'}(z) \le (\nu' - \nu) + z\, I_\nu'(z)/I_\nu(z)$.*

(iv) *For fixed $z > 0$, $I_\nu(z)$ is nonincreasing in $\nu \ge 0$.*

(v) $I_\nu(z) \le \dfrac{(z/2)^\nu}{\Gamma(\nu+1)} e^{z^2/4}$.

Proof. (i) $I_\nu(z)/z^\nu = 2^{-\nu}\sum_k (z^2/4)^k/(k!\,\Gamma(\nu+k+1))$ has positive coefficients. (v) follows from the same series and $\Gamma(\nu+k+1) \ge \Gamma(\nu+1)$. For (ii) and (iii) we use the Weierstrass product $I_\nu(z) = \frac{(z/2)^\nu}{\Gamma(\nu+1)} \prod_{k\ge1} \left(1 + z^2/j_{\nu,k}^2\right)$, where $0 < j_{\nu,1} < j_{\nu,2} < \dots$ are the positive zeros of J_ν [33, §15.41]. It gives

$$\frac{z I_\nu'(z)}{I_\nu(z)} = \nu + \sum_{k\ge1} \frac{2z^2}{j_{\nu,k}^2 + z^2},$$

each summand increasing in z, which is the first part of (ii). For the second, $\frac{d}{dy}\log\frac{I_\nu(cy)}{I_\nu(y)} = \frac{1}{y}\left(\frac{cyI_\nu'(cy)}{I_\nu(cy)} - \frac{yI_\nu'(y)}{I_\nu(y)}\right) \le 0$. For (iii), the zeros $j_{\nu,k}$ increase with ν [33, §15.6], so each summand for ν' is at most the one for ν. (iv) is a theorem of Cochran [10]. □

Lemma 9.2. (i) *$0 < g(y) < 1$ for all $y > 0$, and $g(y) \to 1$ as $y \to \infty$.*
(ii) *ϕ is positive and decreasing on $[3,\infty)$; in particular g is increasing on $[3,\infty)$, and for $0 < c \le 1$ the ratio $g(cy)/g(y)$ is nondecreasing on $\{y:\ cy \ge 3\}$.*
(iii) *$y\,\phi(y) \le 29$ for $y \ge 30$; hence $g'(y) \le 29/y^2$ there.*
(iv) *$\sqrt{2\pi y}\,e^{-y} I_\nu(y) \le 1$ for every $\nu \ge \frac{15}{2}$ and $y > 0$.*

Proof. For half-integral order there is the closed form [33, §3.71]

$$g(y) = p\Big(-\frac{1}{2y}\Big) + e^{-2y} p\Big(\frac{1}{2y}\Big), \qquad p(w) = \sum_{k=0}^{7} \frac{(7+k)!}{k!\,(7-k)!} w^k, \tag{47}$$

which gives $g(y) \to 1$. Write $r = I_{17/2}/I_{15/2}$; the recurrences give $\phi = 8 - y + yr$ and $\phi' = y - 1 - 15r - yr^2$ (we use $I'_\nu = I_{\nu+1} + \frac{\nu}{y} I_\nu$ and $r' = 1 - \frac{16}{y} r - r^2$). On $[3, 60]$ we enclose ϕ' on a cover of the interval by boxes, in ball arithmetic, by the mean value form $\phi'(c) + [-\frac{h}{2}, \frac{h}{2}]\,\phi''(\text{box})$; every enclosure is negative. On $[60,\infty)$ we use (47): with $w = 1/2y$, the main part $w\,p'(-w)/p(-w)$ of ϕ has positive derivative in w on $(0, \frac{1}{120}]$ (checked on 600 subintervals), and the remaining part is $O(y^2 e^{-2y})$ together with its derivative, which is far smaller. This proves that ϕ is decreasing on $[3,\infty)$, and $\phi(60) > 0$ together with $\phi \to 0^+$ gives positivity. Since g increases on $[3,\infty)$ to the limit 1, $g < 1$ there; on $(0, 3]$, (v) of Lemma 9.1 gives $g(y) \le \sqrt{6\pi}\,(3/2)^{15/2} e^{9/4-3}/\Gamma(17/2) < 0.01$. The monotonicity of $g(cy)/g(y)$ is (ii) of Lemma 9.1 with ϕ in place of zI'/I. (iii) on $[30, 60]$ is checked on a grid of step 0.05 using that ϕ is decreasing, and on $[60,\infty)$ from the closed form, where $y\phi(y) \le \frac{1}{2} p'(-w)/p(-w) + O(y^3 e^{-2y}) < 28$. Finally (iv) follows from (i) and Lemma 9.1(iv). □

The following monotonicity in x is what makes every bound of Section 10, proved at the smallest admissible x, valid for all larger x.

Lemma 9.3. *Let $x = \sqrt{D}/m$ and let $\mathcal{E}(D,m)$ be as in (41). For a constant $0 < c \le \pi\sqrt{t_0}$, each of the following ratios is a nonincreasing function of $x > 0$:*

$$\frac{D^{-15/4} I_{15/2}(cx)}{\mathcal{E}(D,m)}, \qquad \frac{D^{-4} I_8(cx)}{\mathcal{E}(D,m)}, \qquad \frac{D^{-17/4} I_{17/2}(cx)}{\mathcal{E}(D,m)}.$$

Proof. Up to constants, $\mathcal{E} = D^{-15/4} I_{15/2}(c_0 x)$ with $c_0 = \pi\sqrt{t_0} \ge c$, and $D = m^2 x^2$. The first ratio is $I_{15/2}(cx)/I_{15/2}(c_0 x)$, nonincreasing by Lemma 9.1(ii). The logarithmic derivative in x of the second is

$$\frac{1}{x}\Big(-\frac{1}{2} + \frac{cx I'_8(cx)}{I_8(cx)} - \frac{c_0 x I'_{15/2}(c_0 x)}{I_{15/2}(c_0 x)}\Big) \le \frac{1}{x}\Big(-\frac{1}{2} + \frac{1}{2} + \frac{cx I'_{15/2}(cx)}{I_{15/2}(cx)} - \frac{c_0 x I'_{15/2}(c_0 x)}{I_{15/2}(c_0 x)}\Big) \le 0$$

by Lemma 9.1(iii) and (ii). The third is the same with -1 and $+1$. □

Remark 9.4. In Lemma 9.3 the extra power $D^{-1/4}$ of the shadow terms exactly compensates the higher order of the Bessel function. This is the level-two counterpart of the fact, used in [5, 24], that the shadow terms of a mixed mock form grow at most like its principal-part terms.

Lemma 9.5. *For $n \ge -1$, $0 < d_1(n) \le e^{2\pi\sqrt{2(n+1)}}$. More generally $d_1(n) \le P(q)\,q^{-(n+1)}$ for every $0 < q < 1$.*

Proof. Positivity is clear from the product. The second bound holds since $P(q) = \sum d_1(a) q^{a+1}$ has nonnegative coefficients. With $q = e^{-\tau}$, comparison of the sum with an integral gives $\sum_{k\ge1} -\log(1 - e^{-k\tau}) \le \int_0^\infty -\log(1 - e^{-u\tau})\,du = \pi^2/6\tau$, so $\log P(e^{-\tau}) \le 8 \cdot \frac{\pi^2}{6\tau} + 8 \cdot \frac{\pi^2}{12\tau} = \frac{2\pi^2}{\tau}$. Taking $\tau = \pi\sqrt{2/(n+1)}$ gives the first bound. □

10. Bounds for the remaining terms

Throughout this section (m,n,ℓ) is a reduced charge with $m \geq 3$, $x = \sqrt{D}/m \geq x_0 \geq 1$, and all bounds are uniform in ℓ and in $x \geq x_0$. We write $B_\gamma(t;x) = (t/t_0)^{15/4} I_{15/2}(\pi x\sqrt{t}/\gamma)/I_{15/2}(\pi x\sqrt{t_0})$ and $u = e^{-\pi}$. By Lemma 9.1(i),(ii), $B_\gamma(t;x)$ is increasing in t and nonincreasing in x, and $B_\gamma \leq \gamma^{-15/2} B_1$.

10.1. **Decomposition.** Assume Hypothesis R. Then $d^*(m,n,\ell) = \frac{1}{2}(-1)^\ell \sqrt{i}(\mathcal{M} + \mathcal{K}_2 + \mathcal{K}_3)$, and Theorem 8.1 splits this as

$$\frac{d^*(m,n,\ell)}{\mathcal{E}(D,m)} = |P(z)|^2 - \mathcal{X}_{\mathrm{miss}} + \mathcal{X}_{\mathrm{main}} + \mathcal{X}_{rs} + \mathcal{X}_T + \mathcal{X}_{\gamma\geq 2} + \mathcal{X}_{K_2} + \mathcal{X}_{K_3}, \tag{48}$$

where

- $\mathcal{X}_{\mathrm{miss}} = \sum d_1(a) d_1(b) e^{-\pi x S} \cos((a-b)\theta)$ over the pairs $a, b \geq -1$ not in $\mathcal{L}_m$;
- $\mathcal{X}_{\mathrm{main}} = \sum_{(a,b)\in\mathcal{L}_m} d_1(a) d_1(b) \big(\rho_{a,b}(x) - e^{-\pi x S}\big) \cos((a-b)\theta)$;
- $\mathcal{X}_{rs}$ is the contribution of the walls with $rs \geq 2$ to the frame S term with $\gamma = 1$, divided by $\mathcal{E}$;
- $\mathcal{X}_T$ is the frame T term with $\gamma = 1$, divided by $\mathcal{E}$;
- $\mathcal{X}_{\gamma\geq 2}$ collects the terms of $\mathcal{M}$ with $\gamma \geq 2$, and $\mathcal{X}_{K_j}$ the term $\frac{1}{2}(-1)^\ell\sqrt{i}\,\mathcal{K}_j$, divided by $\mathcal{E}$.

We bound each piece in absolute value.

10.2. **Parametrising the walls.**

Lemma 10.1. *The map sending a polar term $(\tilde\nu, \tilde\ell)$ with $0 \leq \tilde\ell < 2m$ and a wall $G \in \mathcal{W}(m, \tilde\nu, \tilde\ell)$ to (r, s, m_G, n_G) is injective. Given (r,s,a,b) in its image, δ is the unique solution of*

$$\sqrt{\delta^2 + \frac{a}{mr^2}} + \sqrt{\delta^2 + \frac{b}{ms^2}} = \frac{1}{rs}, \tag{49}$$

so $t = 4m^2\delta^2$ is a function $t(r,s,a,b)$, and $\ell_G \leq m - a - b$.

Proof. By Lemma 6.1(ii), $U = \sqrt{\delta^2 + a/mr^2}$ and $W = \sqrt{\delta^2 + b/ms^2}$, and $U + W = 1/rs$; the left side of (49) is increasing in δ^2, which gives uniqueness, and $\ell_G = 2mrs(UW + \delta^2)$ is then determined. The form $\phi \circ G$ has coefficients $(n_G, -\ell_G, m_G)$ up to the order of the variables, so it is determined; given (r,s), the matrices G are $\left(\begin{smallmatrix} p_0 + kr & q_0 + ks \\ r & s \end{smallmatrix}\right)$, $k \in \mathbb{Z}$; for a fixed form $\phi \circ G$ the choice of k shifts $\omega_0 = \tilde\ell/2m$ by the integer k, that is, $\tilde\ell$ by $2mk$; the condition $0 \leq \tilde\ell < 2m$ fixes k, and then $\phi = (\phi \circ G) \circ G^{-1}$ is determined. Finally $\ell_G = m - a - b - E_G \leq m - a - b$ by Lemma 6.1(iii). □

10.3. **The leading family.**

Lemma 10.2. *Let $(a,b) \in \mathcal{L}_m$ with $S \geq 1$, and put $c_0 = \frac{k}{m+2}(t/t_0)^{7/2}$ and $\hat g_0 = g(\pi x_0\sqrt{t})/g(\pi x_0\sqrt{t_0})$. Then*

$$\sup_{x\geq x_0} \big|\rho_{a,b}(x) - e^{-\pi x S}\big| \leq \max_{C\in\{c_0,\, c_0\hat g_0\}} \sup_{x\geq x_0} \big|C e^{-\pi\Delta_s x} - e^{-\pi S x}\big|.$$

Proof. From $I_{15/2}(y) = g(y)e^y/\sqrt{2\pi y}$ one gets $\rho_{a,b}(x) = c_0\,\hat g(x)\, e^{-\pi\Delta_s x}$ with $\hat g(x) = g(\pi x\sqrt{t})/g(\pi x\sqrt{t_0})$. Since $t \geq 1$, $\pi x\sqrt{t} \geq 3$, and Lemma 9.2(ii) shows $\hat g_0 \leq \hat g(x) \leq 1$. The expression is affine in $\hat g$, so its modulus is at most the larger of the values at the two ends. □

The supremum of $|Ce^{-\alpha x}-e^{-\beta x}|$ on $[x_0,\infty)$ is attained at x_0 or at the unique critical point $x_*=\log(\beta/C\alpha)/(\beta-\alpha)$ when $x_*>x_0$, and is computed exactly. Summing,

$$|\mathcal{X}_{\mathrm{main}}| \le \sum_{(a,b)\in\mathcal{L}_m,\ S\ge 1} d_1(a)d_1(b)\max_{C}\sup_{x\ge x_0}\left|Ce^{-\pi\Delta_s x}-e^{-\pi S x}\right| =: \epsilon_{\mathrm{main}}(m,x_0), \tag{50}$$

$$|\mathcal{X}_{\mathrm{miss}}| \le \sum_{(a,b)\notin\mathcal{L}_m} d_1(a)d_1(b)u^{x_0 S} =: \epsilon_{\mathrm{miss}}(m,x_0). \tag{51}$$

The pairs not in $\mathcal{L}_m$ have $S\ge m/2+2$ (if $t\le 0$ then $(m+2-S)^2\le 4ab\le(S-2)^2$) or $|a-b|\ge m$ (so $S\ge m$) or $k\le 0$; we sum them exactly for $S\le 4m+10$, and for larger S use $\sum_{a+b=S-2} d_1(a)d_1(b)\le P(q_1)^2 q_1^{-S}$ with $q_1=u^{x_0/2}$, by Lemma 9.5.

For large m a coarser bound is convenient.

Lemma 10.3. *Let $m\ge 3$, $x_0=1$, $1\le S\le m+1$, and let $\eta_\pm$ be as in Lemma 8.2. Assume $S-\eta_-\ge 1/\pi$. Then every $(a,b)\in\mathcal{L}_m$ with $a+b+2=S$ satisfies*

$$\sup_{x\ge 1}\left|\rho_{a,b}(x)-e^{-\pi x S}\right| \le u^S\,\varphi(S,m), \qquad \varphi(S,m)=\epsilon_1+e^{\pi\eta_-}-1+\min(1,\pi\eta_+),$$

where $\epsilon_1=\min\left(1,\ \frac{8S}{m}+\frac{29\pi(S+\eta_+)}{y_t^2 g(y_t)}\right)$ with $y_t=\pi\sqrt{t_0-2mS}$ if $y_t\ge 30$, and $\epsilon_1=1$ otherwise.

Proof. Write $\rho=c_0\hat g\, e^{-\pi x\Delta_s}=c\,e^{\pi x y}e^{-\pi x S}$ with $c=c_0\hat g\le 1$ and $y=S-\Delta_s\in[-\eta_+,\eta_-]$. Then $|ce^{\pi xy}-1|\le c|e^{\pi xy}-1|+(1-c)\le(e^{\pi x\eta_-}-1)+\min(1,\pi x\eta_+)+(1-c)$. First, $1-c\le(1-\frac{k}{m+2})+(1-(t/t_0)^{7/2})+(1-\hat g)$; here $1-\frac{k}{m+2}=\frac{S}{m+2}$, $1-(t/t_0)^{7/2}\le\frac{7}{2}\cdot\frac{2mS}{t_0}\le\frac{7S}{m}$ by Lemma 8.2(i), and, with $z=\pi x\sqrt{t}\ge y_t$ and $z_0=\pi x\sqrt{t_0}$, Lemma 9.2(ii),(iii) give

$$1-\hat g\le\frac{g(z_0)-g(z)}{g(z)}\le\frac{29(z_0-z)}{z^2 g(z)}\le\frac{29\pi x(S+\eta_+)}{\pi^2x^2t\,g(y_t)}\le\frac{29\pi(S+\eta_+)}{y_t^2 g(y_t)}$$

when $y_t\ge 30$; trivially $1-c\le 1$. Multiply by $e^{-\pi xS}$. The function $e^{-\pi xS}(e^{\pi x\eta_-}-1)=e^{-\pi x(S-\eta_-)}-e^{-\pi xS}$ is nonincreasing on $[1,\infty)$ as soon as $S/(S-\eta_-)\le e^{\pi\eta_-}$, which follows from $e^{\pi\eta_-}\ge 1+\pi\eta_-$ and $S-\eta_-\ge 1/\pi$; the function $xe^{-\pi xS}$ is decreasing for $x\ge 1/\pi S$; and the bound for $1-c$ is nonincreasing in x. So the supremum is at most the value at $x=1$. □

10.4. **The other walls at $\gamma=1$.** Dividing (42) restricted to a_S^{rest}, and the frame T term, by $\mathcal{E}$, and using $|W_{1,\delta}|=(2m)^{-1/2}$, Lemma 10.1 and the monotonicity of B_1,

$$|\mathcal{X}_{rs}| \le \frac{1}{m+2}\sum_{\substack{r,s\le 2m,\ rs\ge 2\\ a,b\ge -1}}(m-a-b)\,d_1(a)d_1(b)\,B_1\big(t(r,s,a,b);x_0\big) =: \epsilon_{rs}, \tag{52}$$

$$|\mathcal{X}_T| \le \frac{16}{m+2}\sum_{\substack{r,s\le 2m\\ r\text{ or }s\text{ even}}}\ \sum_{a,b\in\frac12\mathbb{Z}}(m-a-b)\,w(r,s,a,b)\,B_1\big(t(r,s,a,b);x_0\big) =: \epsilon_T, \tag{53}$$

where $w=d_1(a)d_1(2b)$ if r is even and $w=d_1(2a)d_1(b)$ if s is even. The ranges are finite by Lemma 6.1(v): $a\le m/s^2$, $b\le m/r^2$. Only quadruples for which (49) has a solution occur; in the certificate an upper bound for δ^2 is found by bisection and then certified in ball arithmetic, and a quadruple is discarded only when the left side of (49) exceeds $1/rs$ already at the least admissible δ^2, again certified.

For $m\ge 101$ we use instead the following bound.

Lemma 10.4. *For $m \geq 101$, with $c = \frac{m-1}{4(m+2)}$, $q = e^{-\pi c}$, $A_1 = \frac{m}{2} - 3 - \frac{2}{m}$ and $g_0 = g(\pi\sqrt{t_0})$,*

$$|\mathcal{X}_{rs}| + |\mathcal{X}_T| \leq \frac{(2m+1)^2\big(P(q)^2 + 64\sqrt{q}\,P(q)P(\sqrt{q})\big)e^{-\pi A_1/2}}{g_0}.$$

Proof. For a wall with $rs \geq 2$, Lemma 6.1(vi) gives $\sqrt{t} \leq \frac{m}{2} + 5$, and $\sqrt{t_0} \geq m + 2 - \frac{2}{m}$, so $\Delta_s = \sqrt{t_0} - \sqrt{t} \geq A_1$. Also $\Delta_s \geq S(2m - S)/(2(m+2))$ with $S = a + b + 2 \leq m + 1$ (by Lemma 6.1(ii), $a + b \leq m(r^2U^2 + s^2W^2)$, and on the segment $U + W = 1/rs$ this convex function is at most $m\max(r^{-2}, s^{-2}) \leq m$; $a + b = m$ is excluded since $\ell_G \leq m - a - b$ must be positive), hence $\Delta_s \geq 2cS$. So $e^{-\pi x\Delta_s} \leq e^{-\pi A_1/2}q^S$. As in Lemma 10.2, $B_1(t;x) \leq (t/t_0)^{7/2}e^{-\pi x\Delta_s}g(\pi x\sqrt{t})/g(\pi x\sqrt{t_0}) \leq e^{-\pi x\Delta_s}/g_0$. With $m - a - b \leq m + 2$, the sum over (a,b) of $d_1(a)d_1(b)q^S$ is at most $P(q)^2$, the sum of $d_1(a)d_1(2b)q^{a+b+2}$ over $a \in \mathbb{Z}$, $b \in \frac{1}{2}\mathbb{Z}$ is at most $\sqrt{q}\,P(q)P(\sqrt{q})$, and there are at most $(2m)^2$ pairs (r,s). The factor 64 accounts for the factor 16, the two parity types and a factor 2 of room. □

10.5. **The terms with $\gamma \geq 2$.** Let $N_F(\gamma)$ be the number of δ with $(\gamma,\delta) \in \mathcal{C}_\gamma$ and frame F; then $N_I + N_T + N_S \leq 2\gamma$. Each term of $\mathcal{M}$ with $\gamma \geq 2$, divided by $\mathcal{E}$, is at most $\frac{16}{m+2}|a_F| \cdot \sqrt{2m}\,|W| \cdot \gamma^{-1}B_\gamma(t;x)$, and $\sqrt{2m}|W| \leq \sqrt{2\gcd(m,\gamma)}$ by Lemma 7.1. Bounding $|a_S|$ and $|a_I|$ by the sums of the absolute values of their walls,

$$\begin{aligned}|\mathcal{X}_{\gamma\geq 2}| \leq \sum_{\gamma=2}^{G} \frac{\sqrt{2\gcd(m,\gamma)}}{\gamma}\big(N_S(\gamma)\Sigma_S(\gamma) + (N_I + N_T)(\gamma)\,\Sigma_T(\gamma)\big)\\ + \frac{2\sqrt{2}\,(\Sigma_S(1) + \Sigma_T(1))}{6G^6} =: \epsilon_{\gamma\geq 2},\end{aligned} \tag{54}$$

where $\Sigma_S(\gamma)$ and $\Sigma_T(\gamma)$ are the sums (52) (including the leading family) and (53) with B_1 replaced by B_γ. The tail uses $B_\gamma \leq \gamma^{-15/2}B_1$, $N \leq 2\gamma$, $\gcd(m,\gamma) \leq \gamma$ and $\sum_{\gamma>G}\gamma^{-7} \leq G^{-6}/6$. We take $G = 12$.

Lemma 10.5. *For $m \geq 101$, with q, g_0 as in Lemma 10.4,*

$$|\mathcal{X}_{\gamma\geq 2}| \leq 2^9\big(\zeta(7) - 1\big)\sqrt{2}\,e^{-\pi\sqrt{t_0}/2}\,\frac{(2m+1)^2\big(P(q)^2 + 64\sqrt{q}\,P(q)P(\sqrt{q})\big)}{g_0}.$$

Proof. By Lemma 9.1(i), $I(z/\gamma) \leq (2/\gamma)^{15/2}I(z/2)$ for $\gamma \geq 2$. By Lemma 9.2(i), $I(z/2) \leq e^{z/2}/\sqrt{\pi z}$, and $I(z_0) = g(z_0)e^{z_0}/\sqrt{2\pi z_0}$, so $(t/t_0)^{15/4}I(z/2)/I(z_0) \leq \sqrt{2}e^{-z_0/2}e^{-(z_0-z)/2}/g_0 \leq \sqrt{2}e^{-\pi\sqrt{t_0}/2}q^S/g_0$, using $(t/t_0)^{15/4}(t_0/t)^{1/4} \leq 1$ and $\Delta_s \geq 2cS$ as before. Finally $\sum_{\gamma\geq 2} 2\gamma \cdot \sqrt{2\gamma}\,\gamma^{-1}(2/\gamma)^{15/2} = 2^9\sum_{\gamma\geq 2}\gamma^{-7}$. □

10.6. **The shadow terms.** By (40) and Lemma 7.1, with $\psi_1 = (2m)^{-1/2}$ and $\psi_\gamma = \sqrt{\gcd(m,\gamma)/m}$, and $D_0 = m^2x_0^2$,

$$|\mathcal{X}_{K_2}| \leq \sum_{\gamma\geq 1}\sum_{F} \frac{N_F(\gamma)}{2} \cdot \frac{\sqrt{2m}\,|c_F|\,\psi_\gamma}{\sqrt{\gamma}}\Big(\frac{4m\mu_F}{D_0}\Big)^4 \frac{I_{15/2}\big(2\pi x_0\sqrt{\mu_F m}/\gamma\big)}{\mathcal{E}(D_0,m)}, \tag{55}$$

$$|\mathcal{X}_{K_3}| \leq \sum_{\gamma\geq 1}\sum_{F} \frac{N_F(\gamma)}{2} \cdot \frac{|c_F|\,\psi_\gamma}{2\pi\gamma^2}\Big(\frac{4m\mu_F}{D_0}\Big)^{\frac{17}{4}} \cdot 2\sqrt{\frac{\mu_F}{m}} \cdot \frac{4\pi^2m^2\gamma^2}{3} \cdot \frac{I_{15/2}\big(2\pi x_0\sqrt{\mu_F m}/\gamma\big)}{\mathcal{E}(D_0,m)}. \tag{56}$$

Indeed, the arguments of I_8 and $I_{17/2}$ are cx with $c \leq 2\pi\sqrt{m} \leq \pi\sqrt{t_0}$, so by Lemma 9.3 each term divided by $\mathcal{E}$ is largest at $x = x_0$; there $I_8, I_{17/2} \leq I_{15/2}$ by Lemma 9.1(iv), the factor $(1 - mu^2/\mu_F)^{17/4}$ is at most 1, $I_{17/2}$ is increasing in its argument, and the interval of

integration has length $2\sqrt{\mu_F/m}$. The certificate sums $\gamma \le G = 12$ exactly and bounds the rest using $I_{15/2}(z/\gamma) \le \gamma^{-15/2} I_{15/2}(z)$, $N \le 2\gamma$ and $\psi_\gamma \le 1$, with $\sum_{\gamma>G} \gamma^{-7} \le G^{-6}/6$ for (55) and $\sum_{\gamma>G} \gamma^{-13/2} \le \frac{2}{11} G^{-11/2}$ for (56).

Lemma 10.6. *For $m \ge 101$, with $z_0 = \pi\sqrt{t_0}$, $z_1 = 2\pi\sqrt{m}$ and $J = \sqrt{z_0/z_1}\, e^{z_1 - z_0}/g(z_0)$,*

$$|\mathcal{X}_{K_2}| \le \frac{\sqrt{2m}\,\frac{17}{16} e^{2\pi\sqrt{2m+2}} (4/m)^4 \zeta(7) J}{\frac{\pi}{\sqrt{2m}}\frac{m+2}{16}}, \qquad |\mathcal{X}_{K_3}| \le \frac{\frac{17}{16} e^{2\pi\sqrt{2m+2}}\frac{1}{2\pi}(4/m)^{17/4}\frac{2}{\sqrt{m}}\frac{4\pi^2 m^2}{3}\zeta(\frac{13}{2}) J}{\frac{\pi}{\sqrt{2m}}\frac{m+2}{16}}.$$

Proof. Take $x_0 = 1$ in (55), (56). Use $|c_F| \le \hat{f}(m) \le e^{2\pi\sqrt{2m+2}}$ (Lemma 9.5), $\mu_F \le 1$, $\psi_\gamma \le 1$, $N \le 2\gamma$, $I_{15/2}(z_1/\gamma) \le \gamma^{-15/2} I_{15/2}(z_1)$, and $\mathcal{E}(m^2, m) \ge \frac{\pi}{\sqrt{2m}}\frac{m+2}{16} I_{15/2}(z_0)$. Finally $I_{15/2}(z_1)/I_{15/2}(z_0) \le J$ by Lemma 9.2(i). □

11. Proof of Theorem C

Let $\epsilon(m, x_0) = \epsilon_{\text{main}} + \epsilon_{\text{miss}} + \epsilon_{rs} + \epsilon_T + \epsilon_{\gamma\ge 2} + \epsilon_{K_2} + \epsilon_{K_3}$ be the sum of the bounds of Section 10, where ϵ_{K_j} denote the right sides of (55) and (56) with the tails included.

Theorem 11.1. *Put $x_0(m) = \sqrt{500}/m$ for $3 \le m \le 7$ and $x_0(m) = 1$ for $8 \le m \le 100$. Then*

$$\epsilon\big(m, x_0(m)\big) \le 0.347 \qquad (3 \le m \le 100),$$

with the largest value at $m = 8$; for $3 \le m \le 7$ one has $\epsilon \le 0.0025$.

Proof. Every term of $\epsilon(m, x_0)$ is a finite sum of explicit quantities, which we evaluate in ball arithmetic with 96 bits [21]. Bessel functions are evaluated by Arb's certified routines, the suprema in (50) by the exact critical points, and the solutions of (49) are enclosed as described after (53). The values are listed in Appendix C and plotted in Figure 4; the computation takes about twelve minutes. □

Theorem 11.2. *For every $m \ge 101$, $\epsilon(m, 1) \le 0.3382$.*

Proof. Leading family and missing pairs. Group the pairs by S. For $1 \le S \le m+1$, Lemma 8.2 gives $S - \eta_- = S(2m - S)/(2m + 4) \ge (2m - 1)/(2m + 4) > 1/\pi$, so Lemma 10.3 applies, and the pairs with $a + b + 2 = S$ contribute at most $p_2(S) u^S \varphi(S, m)$, where $p_2(S) = \sum_{a+b=S-2} d_1(a) d_1(b)$. Missing pairs have $S \ge m/2 + 2$ and contribute at most $p_2(S) u^S$. For $S > 300$ we use, for the pairs in $\mathcal{L}_m$, $|\rho_{a,b} - e^{-\pi x S}| \le e^{-\pi\Delta_s} + e^{-\pi S} \le 2e^{-\pi S(m-1)/(2m+4)} \le 2e^{-0.485\pi S}$ (Lemma 8.2(iv)), and $p_2(S) \le P(q)^2 q^{-S}$ with $q = e^{-0.4\pi}$, so these terms add up to at most $2P(q)^2 \varrho^{301}/(1 - \varrho)$ with $\varrho = e^{-0.485\pi}/q$. Evaluated in ball arithmetic, the total is at most 0.33815 for every $101 \le m \le 300$, the largest value being at $m = 101$.

For $m > 300$ we compare with $m = 300$. The bound η_- decreases in m; for $S \le 3$ one checks $\eta_+ = 0$ (for $S = 3$ this is $\sqrt{1 + 4/m} + \sqrt{1 - 2/m} \ge 2$, true for $m \ge 3$), and for $S \ge 4$ the function $\sqrt{m(m+4)} - \sqrt{m(m+4-2S)} = 2S/\big(\sqrt{1 + 4/m} + \sqrt{1 + (4 - 2S)/m}\big)$ decreases in m; $y_t = \pi\sqrt{m(m+4-2S)}$ increases, and $8S/m$ decreases. Hence $\varphi(S, m) \le \varphi(S, 300)$ whenever $2S \le 304$. For $152 < S \le 300$ the term $\varphi(S, 300)$ has $\epsilon_1 = 1$ and $\min(1, \pi\eta_+) = 1$, so again $\varphi(S, m) \le \varphi(S, 300)$. The missing pairs for $m > 300$ have $S \ge m/2 + 2 > 152$, and their terms $p_2(S) u^S$ with $S \le 300$ are counted at $m = 300$; those with $S > 300$ are in the tail. So the bound at $m = 300$, which is 0.1090, holds for all $m > 300$.

The other pieces. By Lemmas 10.4, 10.5 and 10.6, with $q = q(101)$ (since c increases with m, the bound with $q(101)$ holds for all $m \ge 101$),

$$|\mathcal{X}_{rs}| + |\mathcal{X}_T| \le 3.02 \cdot 10^{-8}, \quad |\mathcal{X}_{\gamma\ge 2}| \le 3 \cdot 10^{-45}, \quad |\mathcal{X}_{K_2}| \le 6 \cdot 10^{-79}, \quad |\mathcal{X}_{K_3}| \le 7 \cdot 10^{-77}$$

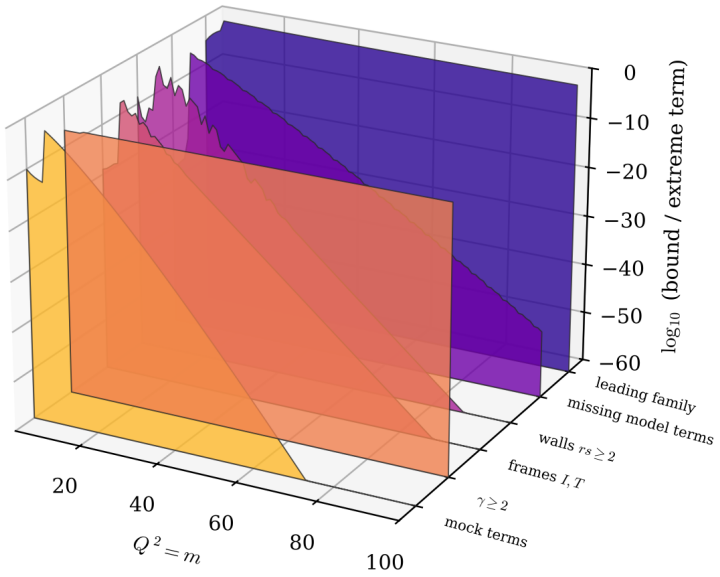


FIGURE 4. The pieces of the certificate for $3 \le Q^2 \le 100$, on a logarithmic scale and divided by the extreme term $\mathcal{E}$: shadow terms, terms with $\gamma \ge 2$, the frames I, T, the walls with $rs \ge 2$, the missing model terms and the error of the leading family. Only the last one matters numerically; the others are below 10^{-2} at $Q^2 = 8$ and fall off quickly.

at $m = 101$. Each bound decreases in m: in Lemma 10.4 the factor $(2m+1)^2 e^{-\pi m/4}$ decreases for $m \ge 101$ and g_0 increases; in Lemma 10.5 the factor $e^{-\pi\sqrt{t_0}/2}$ dominates; in Lemma 10.6 the exponent $2\pi\sqrt{2m+2} + z_1 - z_0 \le 2\pi\sqrt{2m+2} + 2\pi\sqrt{m} - \pi m$ decreases. □

Proof of Theorem C. By Proposition 5.6 it suffices to prove $d^*(m,n,\ell) > 0$ for reduced charges. For $m = 1, 2$ this is Theorem A. Let $m \ge 3$. If $m \le 7$ and $D \le 500$, it is Theorem B. Otherwise $x = \sqrt{D}/m \ge x_0(m)$, with $x_0(m)$ as in Theorem 11.1 for $m \le 100$ and $x_0 = 1$ for $m \ge 101$. By (48) and the bounds of Section 10, which are uniform in $x \ge x_0$ and in ℓ,

$$\frac{d^*(m,n,\ell)}{\mathcal{E}(D,m)} \;\ge\; |P(z)|^2 - \epsilon(m,x_0) \;\ge\; 0.53816 - 0.347 > 0$$

by Proposition 8.3 (note $|z| = e^{-\pi x} \le e^{-\pi}$) and Theorems 11.1 and 11.2. Since $\mathcal{E}(D,m) > 0$, $d^*(m,n,\ell) > 0$. □

Corollary 11.3. *Under Hypothesis* R*, for every reduced charge with* $m \ge 3$*,*

$$0.19\,\mathcal{E}(D,m) \;\le\; d^*(m,n,\ell) \;\le\; 2.51\,\mathcal{E}(D,m).$$

In particular $\log d^*(Q,P) = \pi\sqrt{1+4/Q^2}\,\sqrt{D} - 4\log D + O_{Q^2}(1)$ *as* $D \to \infty$ *with* Q^2 *fixed, and* $\log d^* = \pi\sqrt{Q^2P^2 - (Q\cdot P)^2} + O(\log D)$ *when* $Q^2 \to \infty$ *with* $\sqrt{D}/Q^2$ *bounded, the Bekenstein–Hawking entropy of the black hole.*

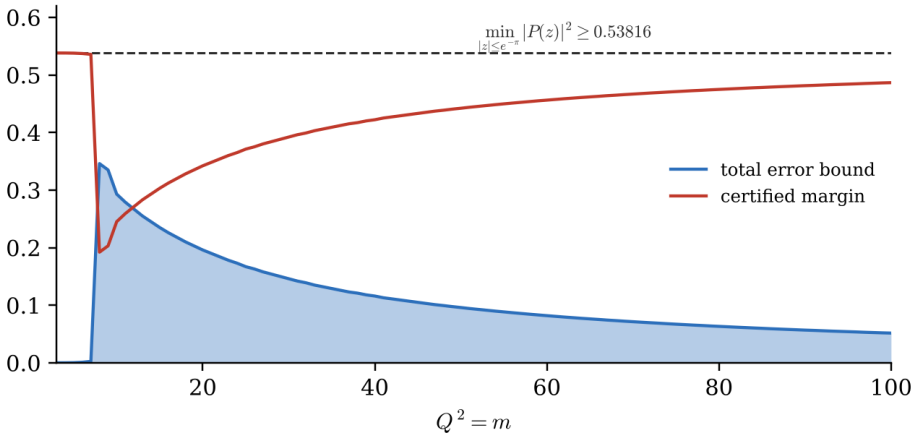


FIGURE 5. The total error bound $\epsilon(m, x_0(m))$ and the margin $0.53816 - \epsilon$ for $3 \le Q^2 \le 100$. The margin is smallest at $Q^2 = 8$, where it is 0.19.

Proof. The lower bound is the proof above. For the upper bound, $|P(z)|^2 \le P(e^{-\pi})^2 \le 2.156$ and $\epsilon \le 0.347$. The asymptotics follow from (41), $I_\nu(y) \sim e^y/\sqrt{2\pi y}$, and $\pi\sqrt{D}\,(\sqrt{1+4/m} - 1) \le 2\pi\sqrt{D}/m$. □

Remark 11.4. The minimum 0.5389 of Proposition 8.3 is reached in the limit: as $m \to \infty$ along charges with $x = 1$ and $\ell = m$ odd, $z = -e^{-\pi}$, and the bounds in the proof of Theorem 11.2 tend to 0, since $\varphi(S, m) \to 0$ for each fixed S. So, under Hypothesis R, $d^*/\mathcal{E}$ comes arbitrarily close to 0.5389, and the factor of the leading family cannot be replaced by a single term; the dominance argument of [24], with one dominant term, does not carry over to level two without the resummation of Theorem 8.1. For moderate charges the picture is already accurate: at $Q^2 = 3$, $P^2/2 = 10$, $Q \cdot P = 3$ the exact index gives $d^*/\mathcal{E} = 0.9485$ while $|P(z)|^2 = 0.9910$, and at $Q^2 = 4$, $P^2/2 = 58$, $Q \cdot P = 4$ the two numbers are 0.99990 and 0.99999.

12. VALUES AND COMPARISON WITH THE PRODUCT FORMULA

The first single-centred indices given by (13) and (25) are listed in Table 1. We expanded the product (3) in exact rational arithmetic and compared it with (12) and (20) on every coefficient $x^e y^\ell$ with $e \le 49$ and $|\ell| \le 18$; the two sides agree. Every entry of Table 1 that lies in Sen's Table 2 [29] coincides with it: at $Q^2 = 1$ the entries 360, 2164, 4352, 18944, 36024, 125860 and the value 2164 at $(P^2, Q \cdot P) = (6, 2)$, and at $Q^2 = 2$ the entries 198144, 67008, 6912 of the row $P^2 = 4$ and 1580672, 671744, 101376, 4352 of the row $P^2 = 6$.[11] Figure 6 shows $\log_{10} d^*$ on the two rows.

Remark 12.1. At $Q^2 = 3$, where $\hat{f}(3) = 1122$, positivity holds for $D \le 800$ by Theorem B and for all D under Hypothesis R by Theorem C. An unconditional proof along the lines of Sections 3 and 4 would need the closed form of φ_3. The coefficients of the product formula that we computed (all $x^e y^\ell$ with $e < 60$) are matched by $g(\frac{\sigma}{2})\varphi_3 = 1122\,\varphi_{2,3}^{\mathrm{opt}} + J_3 = 1122\,\mathcal{A}_{2,3} - 1122\,\mathcal{H}|V_3 + J_3$, with $\varphi_{2,3}^{\mathrm{opt}}$ the form of [13, (9.12)–(9.13)] and J_3 a holomorphic Jacobi form

[11] In these entries the charge lies in the cone (3.9) of [29], and the index of the full partition function equals the single-centred index. The entries of Sen's table with $D \le 0$, such as -2 at $(Q^2, P^2, Q \cdot P) = (1, 2, 2)$ or -16 at $(2, 6, 4)$, are outside the scope of the theorem, as they should be.

$Q^2=1$		$Q^2=2$					
D	d^*	D	d^* (ℓ odd)	D	d^* ($\ell\equiv 0$)	D	d^* ($\ell\equiv 2$)
1	360	3	4352	4	18944	4	6912
2	2164	7	67008	8	198144	8	101376
3	4352	11	671744	12	1580672	12	992256
4	18944	15	5189696	16	10586880	16	7550400
5	36024	19	33581312	20	62035968	20	48253952
6	125860	23	190095232	24	327011328	24	270195712
7	233080	27	968248576	28	1578794128	28	1362534400
8	705536	31	4522209472	32	7075522048	32	6304638848

TABLE 1. The first single-centred indices on the rows $Q^2=1$ and $Q^2=2$, as functions of $D=Q^2P^2-(Q\cdot P)^2$ and, at $Q^2=2$, of $\ell=Q\cdot P$ modulo 4.

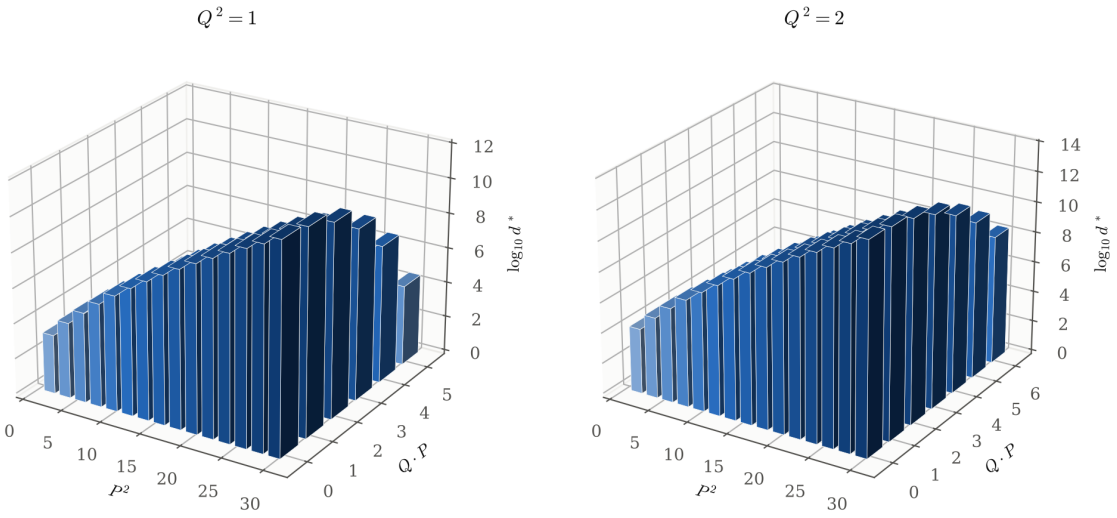


FIGURE 6. $\log_{10} d^*(Q,P)$ for $Q^2=1$ (left) and $Q^2=2$ (right), $2\le P^2\le 30$ and $0\le Q\cdot P\le \min(6,P^2)$ with $Q^2P^2>(Q\cdot P)^2$. Every bar is positive, as Theorems 3.3 and 4.8 assert.

of weight two and index three built from A, B and Eisenstein series at σ and $\sigma/2$. Proving this identity needs the analogue of Proposition 4.1 at index four, and the signs of the theta components of J_3 would then have to be determined.

13. WHAT REMAINS

The proof of Theorem C uses Hypothesis R and nothing else that is unproved. Three routes would remove it.

(1) *The expansion (H).* The method of Bringmann and Manschot [5] writes a mixed mock Jacobi form as a combination of Poincaré series and of the image of the shadow under a non-holomorphic Eichler integral, and proves that the difference is a cusp form of positive weight that vanishes for weight reasons. At level two the Poincaré series have to be summed over the three frames, and the vanishing argument has to be made for $\Gamma^0(2)$; the space of holomorphic cusp forms that could intervene has weight $6+\frac{1}{2}$ in each

theta component, and its vanishing is the one point that needs an idea. The agreement to 21 digits at $Q^2 = 1, 2$ (Section 7.4) shows that no such cusp form is present there.

(2) *The polar data (P0), (P∞).* Cardoso, Nampuri and Rosselló derive the continued-fraction formula [9, (2.43)] by tracking the poles of $1/\widetilde{\Phi}_k$ across the walls of marginal stability. The same contour argument, applied to φ_m^F transformed by $\sigma \mapsto -1/\sigma$, would give (P0); (P∞) is its restriction to the walls fixed by the half shift.

(3) *Exact computation.* Each further unit of Q^2 for which the remainder J_m has a closed form, as at $Q^2 = 1, 2$, removes the hypothesis on that row. Theorem 11.1 shows that the margin for $3 \le Q^2 \le 7$ is above 0.53 as soon as $D > 500$, so on these rows the analytic picture of Section 8 is already in force at moderate discriminant.

The method itself is not special to $N = 2$. For the $\mathbb{Z}_N$ CHL models with $N = 3, 5, 7$ the leading family of walls again resums, now to $|1/\eta(\tau)^{k+2}\eta(N\tau)^{k+2}|^2$ evaluated at $e^{-2\pi x/N+i\theta}$ with $k + 2 = 24/(N + 1)$, and positivity reduces in the same way to a lower bound for this modulus and to an expansion of type (H) at level N.

Declarations

Data availability. The programs and data that support the findings of this study are openly available in the archive [3], at `https://doi.org/10.5281/zenodo.24094932`. They comprise the exact computation of Appendix A, the interval certificates of Appendix B, the checks of Section 7.4, the programs that verify Propositions 3.1, 4.1 and 4.2 and the values of Table 1, and the scripts that draw the figures.

Funding. No funding was received for conducting this study.

Conflict of interest. The author has no conflicts of interest to declare that are relevant to the content of this article.

Ethics statement. This study involves no human participants, animals or personal data; no ethical approval was required.

Appendix A. The exact computation

The program `chl_coefficients.py` of the archive [3] proves Theorem B in exact integer arithmetic, in five steps.

(B1) *Orbits.* A charge (m, n, ℓ) corresponds to the binary quadratic form $f(s, t) = ms^2 + 2\ell st + 2nt^2$ of discriminant $-8D$, and $\Gamma_1(2)$ acts by $f \mapsto f \circ g$. The orbits with given D are enumerated through the Gauss-reduced forms of discriminant $-8D$, each with a class of bottom rows modulo 2, taken modulo the automorphs of the reduced form.

(B2) *A representative in $\mathcal{A}$.* In every orbit the program finds a charge with $0 \le \ell < 2m$ satisfying (10), which it tests in the form $2(jm+\ell m_1)^2+m_1^2\cdot 2D \ge 2m^2$ for $m_1 \in 2\mathbb{Z}\setminus\{0\}$ and odd j with $m_1 \mid (j^2 - 1)/4$, the range of (m_1, j) being finite because the left side grows.

(B3) *Coefficients.* The coefficients $c_{\mathcal{R}}(m, n, \ell)$ are computed from (3) by the recursion for the exponential of (6), with polynomial products in FLINT [20]. By Lemma 5.3(ii) each charge is read in the cheaper of its two orientations.

(B4) *Signs.* For every orbit, $d^* = (-1)^{\ell+1}c_{\mathcal{R}}(m, n, \ell)$ is computed for every representative found in the box, the values are compared, and the sign is checked. At $m = 1, 2$ the values are compared with Propositions 3.1 and 4.2.

(B5) *Least representatives.* On the whole range, every charge of least Q^2 in its orbit is checked to lie in $\mathcal{A}$, as Lemma 5.4 predicts.

For $D \le 800$ there are 22972 orbits, with $Q^2 \le 28$ for the least representative; all have $d^* > 0$. The smallest value relative to the extreme term occurs at small D, where x is close to 1; this is the region in which the leading-family error of Lemma 10.2 is largest and which Theorem B covers for $Q^2 \le 7$.

Appendix B. Interval computations

All the numerical statements used in the proofs are certified in ball arithmetic with Arb [21] through python-flint; each program prints its enclosures.

(I1) `pmin.py`: Proposition 8.3. Product truncated at $n = 60$ with an explicit tail enclosure, 4001 points on the half circle, Lipschitz constant $L = 2P(r_0)r_0P'(r_0)$.

(I2) `gmono2.py`: Lemma 9.2. Monotonicity of ϕ on $[3, 60]$ by mean-value enclosures on an adaptive cover; the range $[60, \infty)$ through the closed form (47).

(I3) `cert.py`: Theorem 11.1. For each m the program lists the leading family $\mathcal{L}_m$, the walls with $rs \ge 2$ and the parity walls through their parameters (r, s, a, b), encloses $t(r, s, a, b)$, and evaluates the seven bounds of Section 10. In (53) it uses 32 in place of 16.

(I4) `analytic.py`, `analytic2.py`: Theorem 11.2.

(I5) `polar.py`: the wall sums (31), (32), used also for the checks (c), (d) of Section 7.4.

Appendix C. The certified margins

Table 2 lists, for $3 \le Q^2 \le 100$, the total bound $\epsilon(m, x_0(m))$ of Theorem 11.1 and the margin $0.53816 - \epsilon$.

Table 2. The certified total error $\epsilon(m, x_0(m))$ of Theorem 11.1 and the margin $0.53816 - \epsilon$, for $m = Q^2$.

m	ϵ	margin	m	ϵ	margin	m	ϵ	margin	m	ϵ	margin
3	$6.04 \cdot 10^{-6}$	0.5380	28	0.1540	0.3842	53	0.0913	0.4467	78	0.0648	0.4732
4	$5.47 \cdot 10^{-5}$	0.5379	29	0.1500	0.3883	54	0.0898	0.4482	79	0.0640	0.4740
5	$2.87 \cdot 10^{-4}$	0.5377	30	0.1460	0.3921	55	0.0884	0.4496	80	0.0633	0.4747
6	$9.79 \cdot 10^{-4}$	0.5370	31	0.1420	0.3958	56	0.0870	0.4510	81	0.0626	0.4754
7	0.0024	0.5356	32	0.1390	0.3993	57	0.0857	0.4523	82	0.0619	0.4761
8	0.3460	0.1917	33	0.1350	0.4026	58	0.0844	0.4536	83	0.0612	0.4768
9	0.3350	0.2030	34	0.1320	0.4058	59	0.0831	0.4549	84	0.0605	0.4775
10	0.2930	0.2450	35	0.1290	0.4089	60	0.0819	0.4561	85	0.0599	0.4781
11	0.2790	0.2594	36	0.1260	0.4118	61	0.0807	0.4573	86	0.0592	0.4788
12	0.2670	0.2714	37	0.1230	0.4145	62	0.0795	0.4585	87	0.0586	0.4794
13	0.2550	0.2828	38	0.1210	0.4172	63	0.0784	0.4596	88	0.0580	0.4800
14	0.2450	0.2930	39	0.1180	0.4197	64	0.0773	0.4607	89	0.0574	0.4806
15	0.2350	0.3026	40	0.1160	0.4222	65	0.0763	0.4617	90	0.0568	0.4812
16	0.2260	0.3115	41	0.1130	0.4245	66	0.0753	0.4627	91	0.0563	0.4817
17	0.2180	0.3199	42	0.1110	0.4268	67	0.0742	0.4638	92	0.0557	0.4823
18	0.2100	0.3277	43	0.1090	0.4289	68	0.0733	0.4647	93	0.0551	0.4829
19	0.2030	0.3350	44	0.1070	0.4310	69	0.0723	0.4657	94	0.0546	0.4834
20	0.1960	0.3419	45	0.1050	0.4330	70	0.0714	0.4666	95	0.0541	0.4839
21	0.1900	0.3483	46	0.1030	0.4350	71	0.0705	0.4675	96	0.0536	0.4844
22	0.1840	0.3544	47	0.1010	0.4368	72	0.0696	0.4684	97	0.0530	0.4850
23	0.1780	0.3601	48	0.0994	0.4386	73	0.0688	0.4692	98	0.0525	0.4855
24	0.1730	0.3655	49	0.0976	0.4404	74	0.0679	0.4701	99	0.0520	0.4860
25	0.1670	0.3706	50	0.0960	0.4420	75	0.0671	0.4709	100	0.0516	0.4864
26	0.1630	0.3754	51	0.0943	0.4437	76	0.0663	0.4717			
27	0.1580	0.3799	52	0.0928	0.4452	77	0.0655	0.4725			

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